Cliffford Algebra Approach to the Representability Problem

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I. REPRESENTATIONS OF CLIFFORD ALGEBRAS

The Pauli matrices are

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{I}_2 = |1\rangle\langle 1| + |2\rangle\langle 2| \tag{1}$$

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x = |1\rangle\langle 2| + |2\rangle\langle 1| \tag{2}$$

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_y = i\left(|2\rangle\langle 1| - |1\rangle\langle 2|\right) \tag{3}$$

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z = |1\rangle\langle 1| - |2\rangle\langle 2| \tag{4}$$

Representations of the complex Clifford algebras in terms of the Pauli matrices are given by

$$\mathrm{Cl}\left(2r,\mathbb{C}\right)\simeq M\left(2^{r},\mathbb{C}\right) \tag{5}$$

$$\mathrm{Cl}\left(2r-1,\mathbb{C}\right)\simeq M\left(2^{r-1},\mathbb{C}\right)\oplus M\left(2^{r-1},\mathbb{C}\right) \tag{6}$$

$$M\left(2^{r},\mathbb{C}\right):\mathbb{C}^{2^{r}}\rightarrow\mathbb{C}^{2^{r}} \tag{7}$$

$$\mathbb{C}^{2r}\simeq\overbrace{\mathbb{C}^{2}\otimes\cdots\otimes\mathbb{C}^{2}}^{\text{r times}} \tag{8}$$

$$\dim\left\{\mathrm{Cl}\left(2r,\mathbb{C}\right)\right\}=2^{2r} \tag{9}$$

and we can define annihilation and creation operators belonging to $\bigotimes^{r}M\left(2,\mathbb{C}\right)$, (Jordan-Wigner transformation), as

$$a_{k}=\bigotimes_{1\leq j\leq k-1}\sigma_{z}\otimes a\otimes\bigotimes_{k+1\leq j\leq r}\mathbb{I}_{2}=\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes a\otimes\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}$$

$$1\leq k\leq r, \tag{10}$$

where

$$a=\begin{pmatrix}0&1\\0&0\end{pmatrix}. \tag{11}$$

Hence a Clifford basis is

$$x_{k}=\frac{1}{\sqrt{2}}\left(a_{k}^{\dagger}+a_{k}\right) \tag{12}$$

$$=\frac{1}{\sqrt{2}}\left(\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes a^{\dagger}\otimes\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}+\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes a\otimes\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}\right) \tag{13}$$

$$=\frac{1}{\sqrt{2}}\left(\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes\otimes\left(a^{\dagger}+a\right)\otimes\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}\right) \tag{14}$$

$$=\frac{1}{\sqrt{2}}\left(\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes\begin{pmatrix}0&1\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}\right) \tag{15}$$

$$x_{k+r}=\frac{i}{\sqrt{2}}\left(a_{k}^{\dagger}-ia_{k}\right)=\frac{i}{\sqrt{2}}\left(\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes\left(a^{\dagger}-ia\right)\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}\right) \tag{16}$$

$$=\frac{i}{\sqrt{2}}\left(\overbrace{\sigma_{z}\otimes\cdots\otimes\sigma_{z}}^{j-1}\otimes\begin{pmatrix}0&-i\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_{2}\otimes\cdots\otimes\mathbb{I}_{2}}^{r-j}\right) \tag{17}$$

e.g. $r=1$

$$x_1 = \frac{1}{\sqrt{2}}\left(\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{1-1}\otimes\begin{pmatrix}0&1\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{1-1}\right) = \frac{1}{\sqrt{2}}\left(a^\dagger + a\right) \tag{18}$$

$$x_2 = \frac{i}{\sqrt{2}}\left(\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{1-1}\otimes\left(a^\dagger - ia\right)\otimes\overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{1-1}\right) = \frac{i}{\sqrt{2}}\left(a^\dagger - ia\right) \tag{19}$$

$r=2$

$$x_1 = \frac{1}{\sqrt{2}}\left(\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{1-1}\otimes\begin{pmatrix}0&1\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{2-1}\right) = \frac{1}{\sqrt{2}}\left(a^\dagger\otimes\mathbb{I}_2 + a\otimes\mathbb{I}_2\right) \tag{20}$$

$$x_2 = \frac{1}{\sqrt{2}}\left(\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{2-1}\otimes\begin{pmatrix}0&1\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{2-2}\right) = \frac{1}{\sqrt{2}}\left(\sigma_z\otimes a^\dagger + \sigma_z\otimes a\right) \tag{21}$$

$$x_3 = \frac{i}{\sqrt{2}}\left(\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{1-1}\otimes\begin{pmatrix}0&-i\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{2-1}\right) = \frac{i}{\sqrt{2}}\left(a^\dagger\otimes\mathbb{I}_2 - ia\otimes\mathbb{I}_2\right) \tag{22}$$

$$x_4 = \frac{i}{\sqrt{2}}\left(\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{2-1}\otimes\begin{pmatrix}0&-i\\1&0\end{pmatrix}\otimes\overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{2-2}\right) = \frac{i}{\sqrt{2}}\left(\sigma_z\otimes a^\dagger - i\sigma_z\otimes a\right) \tag{23}$$

1. Examples

a. r=1 then

even This is the algebra generated by the Pauli matrices themselves i.e.

$$\mathrm{Cl}\left(2,\mathbb{C}\right) \simeq M\left(2,\mathbb{C}\right) \tag{24}$$

$$M\left(2,\mathbb{C}\right) : \mathbb{C}^2 \to \mathbb{C}^2 \tag{25}$$

$$a_1 = \overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{1-1}\otimes a\otimes\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{0} = a \tag{26}$$

$$a_1^\dagger = \overbrace{\mathbb{I}_2\otimes\cdots\otimes\mathbb{I}_2}^{1-1}\otimes a^\dagger\otimes\overbrace{\sigma_z\otimes\cdots\otimes\sigma_z}^{0} = a^\dagger. \tag{27}$$

odd

$$\mathrm{Cl}\left(1,\mathbb{C}\right) \simeq M\left(2^0,\mathbb{C}\right)\oplus M\left(2^0,\mathbb{C}\right) \simeq \mathbb{C}\oplus\mathbb{C}, \tag{28}$$

b. r=2 then

even

$$\mathrm{Cl}\,(4,\mathbb{C}) \simeq M\,(4,\mathbb{C}) \tag{29}$$

$$M\,(4,\mathbb{C}) : \mathbb{C}^4 \to \mathbb{C}^4 \tag{30}$$

The creators and annihilators are given by (Jordan-Wigner transformation)

$$a_1 = \overbrace{\mathbb{I}_2 \otimes \cdots \otimes \mathbb{I}_2}^{2-1-0} \otimes a \otimes \overbrace{\sigma_z \otimes \cdots \otimes \sigma_z}^{0} = \mathbb{I}_2 \otimes a \tag{31}$$

$$a_1^\dagger = \overbrace{\mathbb{I}_2 \otimes \cdots \otimes \mathbb{I}_2}^{2-1-0} \otimes a^\dagger \otimes \overbrace{\sigma_z \otimes \cdots \otimes \sigma_z}^{0} = \mathbb{I}_2 \otimes a^\dagger. \tag{32}$$

$$a_2 = \overbrace{\mathbb{I}_2 \otimes \cdots \otimes \mathbb{I}_2}^{2-1-1} \otimes a \otimes \overbrace{\sigma_z \otimes \cdots \otimes \sigma_z}^{1} = a \otimes \sigma_z \tag{33}$$

$$a_2^\dagger = \overbrace{\mathbb{I}_2 \otimes \cdots \otimes \mathbb{I}_2}^{2-1-1} \otimes a^\dagger \otimes \overbrace{\sigma_z \otimes \cdots \otimes \sigma_z}^{1} = a^\dagger \otimes \sigma_z. \tag{34}$$

The dimension of this Fock space is

$$\dim\{\mathcal{H}_{\mathcal{F}}\} = 4 = 1 + 2 + 1 = 2^2 = 2^r \tag{35}$$

and of this Fock algebra is

$$\dim\{\mathcal{F}\} = 16 = \left(2^2\right)^2 = (2^r)^2 = 2^4 = 2^{2r} \tag{36}$$

as the number of degrees of freedom $\dim\{\mathcal{H}^1\} = r = 2$.The scalar product is of rank $= 2$. The dimension of the Clifford algebra is thus 2 and thus represented by matrices of order $2^2 = 4$

odd

$$\mathrm{Cl}\,(3,\mathbb{C}) \simeq M\,(2,\mathbb{C}) \oplus M\,(2,\mathbb{C}) \tag{37}$$

A. Fock Space Representations

A basis for a Fock representation of the Clifford Algebra $\mathrm{Cl}\,(2r,\mathbb{C}) \simeq M\,(2^r,\mathbb{C})$ is

$$I_{\mathcal{F}}, x_{j_1}, x_{j_1}x_{j_2}, \cdots, x_{j_1}\cdots x_{j_{2r}}; 1 \leq j_1 < \cdots < j_{2r} \leq 2r \tag{38}$$

Number of basis elements and thus dimension of $\mathcal{F}$ is

$$2^{2r} = \sum_{0\leq j\leq 2r} \binom{2r}{j}. \tag{39}$$

1. Examples

For $r = 0$ the basis of $\mathcal{F}$ is

$$I_{\mathcal{F}} \tag{40}$$

$$\dim\{\mathcal{F}\} = 2^0 = 1 \tag{41}$$

and

$$\mathcal{F} \cong M\,(1,\mathbb{C}) \cong \mathbb{C}. \tag{42}$$

For $r = 1$ the basis of $\mathcal{F}$ is

$$I_{\mathcal{F}} \tag{43}$$

$$\dim\{\mathcal{F}\} = 2^2 = 4 \tag{44}$$

and

$$\mathcal{F} \cong M\,(2,\mathbb{C})\,. \tag{45}$$

For $r = 2$ the basis of $\mathcal{F}$ is

$$I_{\mathcal{F}}, x_1, x_2, x_3, x_4, x_1x_2, x_1x_3, x_1x_4, x_2x_3, x_2x_4, x_3x_4, x_1x_2x_3, x_1x_2x_4, x_1x_3x_4, x_2x_3x_4, x_1x_2x_3x_4 \tag{46}$$

$$\dim\{\mathcal{F}\} = 2^4 = 16 \tag{47}$$

and

$$\mathcal{F} \cong M\,(4,\mathbb{C})\,. \tag{48}$$

For $r=3$ the basis of $\mathcal{F}$ is

$$I_{\mathcal{F}}, x_1, x_2, x_3, x_4, x_5, x_6, \tag{49}$$

$$x_1x_2, x_1x_3, x_1x_4, x_1x_5, x_1x_6, x_2x_3, x_2x_4, x_2x_5, x_2x_6, x_3x_4, x_3x_5, x_3x_6, x_4x_5, x_4x_6, x_5x_6, \tag{50}$$

$$x_1x_2x_3, x_1x_2x_4, x_1x_2x_5, x_1x_2x_6, x_1x_3x_4, x_1x_3x_5, x_1x_3x_6, x_2x_3x_4, x_2x_3x_5, x_2x_3x_6, x_3x_4x_5, \tag{51}$$

$$x_3x_4x_6, x_4x_5x_6, \tag{52}$$

$$x_1x_2x_3x_4, x_1x_2x_3x_5, x_1x_2x_3x_6, x_1x_2x_4x_6, x_1x_2x_5x_6, x_1x_3x_4x_5, x_1x_3x_4x_6, x_1x_3x_5x_6, \tag{53}$$

$$x_2x_3x_4x_5, x_2x_3x_4x_6, x_2x_3x_5x_6, x_3x_4x_5x_6, \tag{54}$$

$$x_1x_2x_3x_4x_5, x_1x_2x_3x_4x_6, x_1x_2x_3x_5x_6, x_1x_2x_4x_5x_6, x_1x_3x_4x_5x_6, x_2x_3x_4x_5x_6, x_1x_2x_3x_4x_5x_6, \tag{55}$$

$$\dim\{\mathcal{F}\} = 2^6 = 64, \tag{56}$$

which corresponds to

$$\binom{6}{0}+\binom{6}{1}+\binom{6}{2}+\binom{6}{3}+\binom{6}{4}+\binom{6}{5}+\binom{6}{6} \tag{57}$$

$$= 1+6+15+20+15+6+1 = 64 \tag{58}$$

and $\mathcal{F} \cong M(8,\mathbb{C})$.

B. Number Representation

The sets of operators $\{n_j|\, 1 \le j \le r\}, \{h_j|\, 1 \le j \le r\}$ commute amongst themselves and each other. They are all orthogonal projection operators thus their spectrum is $\{0,1\}$. Also

$$n_j |\Psi\rangle = |\Psi\rangle \Leftrightarrow h_j |\Psi\rangle = 0, \tag{59}$$

thus the simultaneous eigenbasis of $\{n_j|\, 1 \le j \le r\}$ determines a basis for $\mathcal{H}_{\mathcal{F}}$ and is given by

$$|\eta_1, \cdots, \eta_r\rangle, \tag{60}$$

where

$$\eta_j = 0,1;\ 1 \le j \le r, \tag{61}$$

showing that

$$\dim\{\mathcal{H}_{\mathcal{F}}\} = 2^r. \tag{62}$$

II. PROPERTIES OF CLIFFORD ALGEBRAS

A. Trace

The multiplicative unit is $I_{\mathcal{F}}$ and has the property

$$\langle I_{\mathcal{F}} \rangle_0 = I_{\mathcal{F}} \tag{63}$$

and

$$\langle \alpha I_{\mathcal{F}} \rangle_0 = \alpha I_{\mathcal{F}} \tag{64}$$

One can introduce a linear functional, tr,

$$\mathrm{tr} : \mathcal{F} \rightarrow \mathbb{C} \tag{65}$$

that has the properties

$$\langle A \rangle_0 = \mathrm{tr}\,\{A\}\, I_{\mathcal{F}} \tag{66}$$

$$\mathrm{tr}\,\{I_{\mathcal{F}}\} = 1 \tag{67}$$

$$\mathrm{tr}\,\{\alpha I_{\mathcal{F}}\} = \alpha \tag{68}$$

$$\mathrm{tr}\,\{A + B\} = \mathrm{tr}\,\{A\} + \mathrm{tr}\,\{B\} \tag{69}$$

$$\mathrm{tr}\,\{AB\} = \mathrm{tr}\,\{BA\} \tag{70}$$

$$\mathrm{tr}\,\{ABC\} = \mathrm{tr}\,\{CAB\} = \mathrm{tr}\,\{BCA\} \tag{71}$$

$$\left\langle A^{\dagger} B \right\rangle_0 = \mathrm{tr}\left\{A^{\dagger} B\right\} I_{\mathcal{F}} \tag{72}$$

$$|A|^2 = \left\langle A^{\dagger} A \right\rangle_0 = \mathrm{tr}\left\{A^{\dagger} A\right\} I_{\mathcal{F}} \tag{73}$$

and

$$\mathrm{tr}\,\{A\} = 2^{-2r}\, \mathrm{Tr}\,\{A\}\,, \tag{74}$$

where Tr is the normal trace on the Fock representation.

The trace is zero on

$$\bigoplus_{1 \le n \le 2r} \mathcal{F}_n \tag{75}$$

as

$$\left\langle \sum_{1 \le n \le 2r} A_n \right\rangle_0 = 0. \tag{76}$$

B. Volume Element

$$\mathbf{I} = x_1 \wedge \cdots \wedge x_{2r} \tag{77}$$

If $r = \infty$ then this definition needs careful analysis.

C. Geometric Product

The geometric product between two general elements is defined as

$$AB = \sum_{0 \leq m,n \leq 2r} A_m B_n, \tag{78}$$

where

$$A_m B_n = \sum_{j=0}^{\min(m,n)} \left\langle A_m B_n \right\rangle_{|m-n|+2j}. \tag{79}$$

E.g. let

$$A = A_0 + A_1 + A_2 \tag{80}$$

$$B = B_0 + B_1 + B_2 + B_3 + B_4 \tag{81}$$

then

$$AB = A_0 \left(B_0 + B_1 + B_2 + B_3 + B_4\right) + A_1 \left(B_0 + B_1 + B_2 + B_3 + B_4\right) + A_2 \left(B_0 + B_1 + B_2 + B_3 + B_4\right). \tag{82}$$

The parts containing A_1 are given by

$$A_1 B_1 = \sum_{j=0}^{1} \left\langle A_1 B_1 \right\rangle_{2j} = \left\langle A_1 B_1 \right\rangle_0 + \left\langle A_1 B_1 \right\rangle_2 \tag{83}$$

$$A_1 B_2 = \sum_{j=0}^{1} \left\langle A_1 B_2 \right\rangle_{1+2j} = \left\langle A_1 B_2 \right\rangle_1 + \left\langle A_1 B_2 \right\rangle_3 \tag{84}$$

$$A_1 B_3 = \sum_{j=0}^{1} \left\langle A_1 B_3 \right\rangle_{2+2j} = \left\langle A_1 B_3 \right\rangle_2 + \left\langle A_1 B_3 \right\rangle_4 \tag{85}$$

$$A_1 B_4 = \sum_{j=0}^{1} \left\langle A_1 B_4 \right\rangle_{3+2j} = \left\langle A_1 B_4 \right\rangle_3 + \left\langle A_1 B_4 \right\rangle_5 \tag{86}$$

Expanding the operators in a basis gives

$$\langle A_1 B_1\rangle_0 = \left\langle \sum_{1\le j,k\le 2r} A_{1j} B_{1k} x_j x_k \right\rangle_0 = \sum_{1\le j\le 2r} A_{1j} B_{1k} x_j^2 \tag{87}$$

$$\langle A_1 B_1\rangle_2 = \left\langle \sum_{1\le j,k\le 2r} A_{1j} B_{1k} x_j x_k \right\rangle_2 = \sum_{1\le j<k\le 2r} \left(A_{1j}B_{1k} - A_{1k}B_{1j}\right) x_j x_k = \sum_{1\le j<k\le 2r} \begin{vmatrix} A_{1j} & A_{1k} \\ B_{1j} & B_{1k} \end{vmatrix} x_j x_k \tag{88}$$

$$\langle A_1 B_2\rangle_1 = \left\langle \frac{1}{2} \sum_{1\le j,k,l\le 2r} A_{1j} B_{2kl} x_j x_k x_l \right\rangle_1 = \frac{1}{2} \sum_{1\le j,k\le 2r} \left(A_{1j}B_{2jk} - A_{1j}B_{2kj}\right) x_j^2 x_k \tag{89}$$

$$\langle A_1 B_2\rangle_3 = \left\langle \frac{1}{2} \sum_{1\le j,k,l\le 2r} A_{1j} B_{2kl} x_j x_k x_l \right\rangle_3 = \sum_{1\le j<k<l\le 2r} \left(A_{1j}B_{2kl} - A_{1k}B_{2jl} + A_{1l}B_{kj}\right) x_j x_k x_l \tag{90}$$

$$\langle A_1 B_3\rangle_2 = \left\langle \frac{1}{6} \sum_{1\le j,k,l,m\le 2r} A_{1j} B_{3klm} x_j x_k x_l x_m \right\rangle_2 \tag{91}$$

$$= \frac{1}{3} \left\{ \sum_{1\le j<k,l\le 2r} \left(A_{1j}B_{3kll} - A_{1j}B_{3lkl} + A_{1l}B_{3ljk}\right) x_j x_k x_l^2 \right\} \tag{92}$$

$$\langle A_1 B_3\rangle_4 = \left\langle \frac{1}{6} \sum_{1\le j,k,l,m\le 2r} A_{1j} B_{3klm} x_j x_k x_l x_m \right\rangle_4 \tag{93}$$

$$= \sum_{1\le j<k<l<m\le 2r} \left(A_{1j}B_{3klm} - A_{1k}B_{3jlm} + A_{1l}B_{3kjm} - A_{1m}B_{3klj}\right) x_j x_k x_l x_m \tag{94}$$

$$\langle A_1 B_4\rangle_3 = \left\langle \frac{1}{24} \sum_{1\le j_1,j_2,j_3,j_4,j_5\le 2r} A_{1j_1} B_{4j_2j_3j_4j_5} x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} \right\rangle_3 = \tag{95}$$

$$\frac{1}{24} \sum_{1\le j_1,j_2,j_3,j_4,j_5\le 2r} A_{1j_1} B_{4j_2j_3j_4j_5} \{\delta_{j_1j_2} x_{j_3}\wedge x_{j_4}\wedge x_{j_5} - \delta_{j_1j_3} x_{j_2}\wedge x_{j_4}\wedge x_{j_5} + \delta_{j_1j_4} x_{j_2}\wedge x_{j_3}\wedge x_{j_5} + \delta_{j_2j_3} x_{j_1}\wedge x \tag{96}$$

$$-\delta_{j_2j_4} x_{j_1}\wedge x_{j_3}\wedge x_{j_5} + \delta_{j_3j_4} x_{j_1}\wedge x_{j_2}\wedge x_{j_5} + \delta_{j_4j_5} x_{j_1}\wedge x_{j_2}\wedge x_{j_3} - \delta_{j_3j_5} x_{j_1}\wedge x_{j_2}\wedge x_{j_4} \tag{97}$$

$$+\delta_{j_2j_5} x_{j_1}\wedge x_{j_3}\wedge x_{j_4} - \delta_{j_1j_5} x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\} \tag{98}$$

$$\langle A_1 B_4\rangle_5 = \left\langle \frac{1}{24} \sum_{1\le j_1,j_2,j_3,j_4,j_5\le 2r} A_{1j_1} B_{4j_2j_3j_4j_5} x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} \right\rangle_5 \tag{99}$$

$$= \frac{1}{24} \sum_{1\le j_1,j_2,j_3,j_4,j_5\le 2r} A_{1j_1} B_{4j_2j_3j_4j_5} x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5} \tag{100}$$

$$
\begin{aligned}
&x_{j_1}x_{j_2}x_{j_3}x_{j_4}x_{j_5} = x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5} \\
&+\Delta_{j_1j_2}x_{j_3}\wedge x_{j_4}\wedge x_{j_5} - \Delta_{j_1j_3}x_{j_2}\wedge x_{j_4}\wedge x_{j_5} + \Delta_{j_1j_4}x_{j_2}\wedge x_{j_3}\wedge x_{j_5} + \Delta_{j_2j_3}x_{j_1}\wedge x_{j_4}\wedge x_{j_5} \\
&-\Delta_{j_2j_4}x_{j_1}\wedge x_{j_3}\wedge x_{j_5} + \Delta_{j_3j_4}x_{j_1}\wedge x_{j_2}\wedge x_{j_5} + \Delta_{j_4j_5}x_{j_1}\wedge x_{j_2}\wedge x_{j_3} - \Delta_{j_3j_5}x_{j_1}\wedge x_{j_2}\wedge x_{j_4} \\
&+\Delta_{j_2j_5}x_{j_1}\wedge x_{j_3}\wedge x_{j_4} - \Delta_{j_1j_5}x_{j_2}\wedge x_{j_3}\wedge x_{j_4} \\
&+\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_3j_4j_5}x_{j_1} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_4j_5}x_{j} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_5}x_{j_3} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_5}x_{j_4} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4}x_{j_5},
\end{aligned} \tag{101}
$$

$$
\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4} = \Delta_{j_1j_2}\Delta_{j_3j_4} - \Delta_{j_1j_3}\Delta_{j_2j_4} + \Delta_{j_1j_4}\Delta_{j_2j_3}. \tag{102}
$$

The parts containing A_2 are given by

$$
A_2B_1 = \sum_{j=0}^{1}\left\langle A_2B_1\right\rangle_{1+2j} = \left\langle A_2B_1\right\rangle_1 + \left\langle A_2B_1\right\rangle_3 \tag{103}
$$

$$
A_2B_2 = \sum_{j=0}^{2}\left\langle A_2B_2\right\rangle_{2j} = \left\langle A_2B_2\right\rangle_0 + \left\langle A_2B_2\right\rangle_2 + \left\langle A_2B_2\right\rangle_4 \tag{104}
$$

$$
A_2B_3 = \sum_{j=0}^{2}\left\langle A_2B_3\right\rangle_{1+2j} = \left\langle A_2B_3\right\rangle_1 + \left\langle A_2B_3\right\rangle_3 + \left\langle A_2B_3\right\rangle_5 \tag{105}
$$

$$
A_2B_4 = \sum_{j=0}^{2}\left\langle A_2B_4\right\rangle_{2+2j} = \left\langle A_2B_4\right\rangle_2 + \left\langle A_2B_4\right\rangle_4 + \left\langle A_2B_4\right\rangle_6, \tag{106}
$$

where

$$
\left\langle A_2B_1\right\rangle_1 = \tag{107}
$$

$$
\left\langle A_2B_1\right\rangle_3 = \tag{108}
$$

$$
\left\langle A_2B_2\right\rangle_0 = \tag{109}
$$

$$
\left\langle A_2B_2\right\rangle_2 = \tag{110}
$$

$$
\left\langle A_2B_2\right\rangle_4 = \tag{111}
$$

$$
\left\langle A_2B_3\right\rangle_1 = \tag{112}
$$

$$
\left\langle A_2B_3\right\rangle_3 = \tag{113}
$$

$$
\left\langle A_2B_3\right\rangle_5 = \tag{114}
$$

$$\langle A_2 B_4 \rangle_2 = \tag{115}$$

$$\langle A_2 B_4 \rangle_4 = \tag{116}$$

$$\langle A_2 B_4 \rangle_6 = \tag{117}$$

D. Blades and Versors

A blade is

$$\mathrm{b}_p = v_1 \wedge \cdots \wedge v_p = \langle v_1 \wedge \cdots \wedge v_p \rangle_p \tag{118}$$

while a versor is

$$\mathrm{v}_p = v_1 \cdots v_p = \sum_{0 \le q \le p} \langle v_1 \cdots v_p \rangle_q \,. \tag{119}$$

E.g.

$$\mathrm{v}_2 = \sum_{1 \le j \le k \le 2r} v_{jk} x_j x_k = \left(\sum_{1 \le j \le 2r} v_{jj} \right) I_{\mathcal{F}} + \sum_{1 \le j < k \le 2r} v_{jk} x_j x_k. \tag{120}$$

E. Inverse

If v_p is invertible its inverse is given by

$$\mathrm{v}_p^{-1} = \frac{\mathrm{v}_p^\dagger}{|\mathrm{v}_p|^2} \tag{121}$$

where $\dagger$ is reversion (composed with complex conjugation)

$$\mathrm{v}_p^\dagger = v_p \cdots v_1 = (-1)^{\frac{p(p-1)}{2}} v_1 \cdots v_p \tag{122}$$

grade involution $*$

$$\mathrm{v}_p^* = (-1)^p \, \mathrm{v}_p \tag{123}$$

the scaler product

$$\left\langle \mathrm{v}_p^\dagger \mathrm{w}_p \right\rangle_0 = v_1 w_1 \cdots v_p w_p \tag{124}$$

and thus the norm

$$|\mathrm{v}_p|^2 = \mathrm{v}_p * \mathrm{v}_p = \left\langle \mathrm{v}_p^\dagger \mathrm{v}_p \right\rangle_0 = v_1^2 \cdots v_p^2. \tag{125}$$

Clifford conjugation

$$\mathrm{v}_p^\ddagger = \mathrm{v}_p^{\dagger *} = (-1)^{\frac{2p+p(p-1)}{2}} v_1 \cdots v_p = (-1)^{\frac{p(p+1)}{2}} v_1 \cdots v_p \tag{126}$$

F. Inner Products

$$A\rfloor B = \sum_{0\leq p,q\leq 2r} \left\langle A_p B_q \right\rangle_{p-q} \tag{127}$$

$$A\lfloor B = \sum_{0\leq p,q\leq 2r} \left\langle A_p B_q \right\rangle_{q-p} \tag{128}$$

In particular

$$a\rfloor A_p = \frac{1}{2}\left(aA_p - (-1)^p A_p a\right) \tag{129}$$

$$a\lfloor A_p = \frac{1}{2}\left(A_p a - (-1)^p aA_p\right) \tag{130}$$

and

$$(a|\, b) = a\rfloor b = \langle ab\rangle_0 \,. \tag{131}$$

Thin dot

$$A\cdot B = \sum_{1\leq p,q\leq 2r} \left\langle A_p B_q \right\rangle_{|p-q|} \tag{132}$$

fat dot

$$A \bullet B = \sum_{0\leq p,q\leq 2r} \left\langle A_p B_q \right\rangle_{|p-q|} \,. \tag{133}$$

and *Trace*

$$\mathrm{Tr}\left\{A\right\} = \langle A\rangle_0 \,. \tag{134}$$

G. Outer Product

$$A\wedge B = \sum_{0\leq p,q\leq 2r} \left\langle A_p B_q \right\rangle_{q+p} \tag{135}$$

In particular

$$A_4\wedge A_4 = \langle A_4 A_4\rangle_8 \tag{136}$$

and

$$A \wedge A = \sum_{0\leq p,q\leq 4} \langle A_p A_q \rangle_{q+p} = \tag{137}$$

$$+ \langle A_0 A_0\rangle_0 + 2\langle A_1 A_0\rangle_1 + 2\langle A_2 A_0\rangle_2 + 2\langle A_3 A_0\rangle_3 + 2\langle A_4 A_0\rangle_4 \tag{138}$$

$$+ 2\langle A_0 A_1\rangle_2 + \langle A_1 A_1\rangle_2 + 2\langle A_2 A_1\rangle_3 + 2\langle A_3 A_1\rangle_4 + 2\langle A_4 A_1\rangle_5 \tag{139}$$

$$+ 2\langle A_0 A_2\rangle_2 + 2\langle A_1 A_2\rangle_3 + \langle A_2 A_2\rangle_4 + 2\langle A_3 A_2\rangle_5 + 2\langle A_4 A_2\rangle_6 \tag{140}$$

$$+ 2\langle A_0 A_3\rangle_3 + 2\langle A_1 A_3\rangle_4 + 2\langle A_2 A_3\rangle_5 + \langle A_3 A_3\rangle_6 + 2\langle A_4 A_3\rangle_7 \tag{141}$$

$$+ 2\langle A_0 A_4\rangle_4 + 2\langle A_1 A_4\rangle_5 + 2\langle A_2 A_4\rangle_6 + 2\langle A_3 A_4\rangle_7 + \langle A_4 A_4\rangle_8 . \tag{142}$$

H. Scalar Product

$$A * B = \left\langle A^\dagger B\right\rangle_0 = \sum_{0\leq p\leq 2r} A_p * B_p = \sum_{0\leq p\leq 2r} A_p^\dagger \rfloor B_p = \sum_{0\leq p\leq 2r} A_p^\dagger \lfloor B_p \tag{143}$$

$$|A|^2 = A * A \text{ (which could be zero, negative or even a number!)} \tag{144}$$

$$A_p * B_q = A_p * B_p \delta_{pq}. \tag{145}$$

One can extend this definition by setting

$$\{A|\, B\} = \mathrm{tr}\,\{A * B\}\,. \tag{146}$$

1. Frames and Matrices

One can introduce orthonormal or biorthoganol bases in $\mathcal{F}$ using these products.

The basis $\{x_j|\, 1 \leq j \leq 2r\}$ is frame as

$$x_1 \wedge \cdots\cdots \wedge x_{2r} \neq 0, \tag{147}$$

which is orthogonal

$$x_j * x_k = \left\langle x_j^\dagger x_k\right\rangle_0 = \left\langle x_j^\dagger x_k\right\rangle_0 = 0 \tag{148}$$

but not normalized as

$$\left\langle x_j^\dagger x_j\right\rangle_0 = \frac{1}{2} I_{\mathcal{F}}. \tag{149}$$

It is easy to construct non orthogonal vectors e.g.

$$v = x_1 + x_2 \tag{150}$$

$$w = 2x_1 - x_2 \tag{151}$$

$$v * w = I_{\mathcal{F}} - \frac{1}{2} I_{\mathcal{F}} = \frac{1}{2} I_{\mathcal{F}}. \tag{152}$$

I. Adjoint

Let linear transformation F

$$F : \mathcal{F}_1 \to \mathcal{F}_1 \tag{153}$$

then it can be naturally extended to

$$F : \mathcal{F} \to \mathcal{F} \tag{154}$$

and is an algebra homomorphism wrt geometric product $\wedge$ i.e.

$$F(A \wedge B) = F(A) \wedge F(B)\ \forall A, B \in \mathcal{F} \tag{155}$$

and if F is an isometry i.e.

$$F(A \rfloor B) = F(A) \rfloor F(B)\, \forall A, B \in \mathcal{F} \tag{156}$$

then

$$F(AB) = F(A)\, F(B)\ \forall A, B \in \mathcal{F}. \tag{157}$$

The converse is also true i.e.

$$F(AB) = F(A)\, F(B)\ \forall A, B \in \mathcal{F} \Rightarrow F(A \rfloor B) = F(A) \rfloor F(B)\, \forall A, B \in \mathcal{F}. \tag{158}$$

The adjoint $\mathcal{F}^{\blacksquare}$ is defined by

$$\mathcal{F}^{\blacksquare}(A) * B = A * F(B)\, \forall B \in \mathcal{F}. \tag{159}$$

Let

$$A_p = a_1 \wedge \cdots \wedge a_p \tag{160}$$

$$B_p = b_1 \wedge \cdots \wedge b_p \tag{161}$$

then

$$A_p * B_p = (-1)^{\frac{p(p-1)}{2}} \left(a_p \wedge \cdots \wedge a_1 \middle| b_1 \wedge \cdots \wedge b_p \right) = (-1)^{\frac{p(p-1)}{2}} \left\langle \left(a_p \wedge \cdots \wedge a_1 \right) \left(b_1 \wedge \cdots \wedge b_p \right) \right\rangle_0 \tag{162}$$

$$= (-1)^{\frac{p(p-1)}{2}} \sum_{\substack{1 \le j_1 < \cdots < j_p \le 2r \\ 1 \le k_1 < \cdots < k_p \le 2r}} \left\langle a_{pj_1} \cdots a_{1j_1} x_{j_p} \wedge \cdots \wedge x_{j_1} b_{pk_p} \cdots b_{1k_1} x_{k_1} \wedge \cdots \wedge x_{k_p} \right\rangle_0 \tag{163}$$

$$= (-1)^{\frac{p(p-1)}{2}} \sum_{\substack{1 \le j_1 < \cdots < j_p \le 2r \\ 1 \le k_1 < \cdots < k_p \le 2r}} \begin{vmatrix} \left(a_{j_1} \middle| b_{k_1} \right) & \cdots & \left(a_{j_1} \middle| b_{k_p} \right) \\ \vdots & \vdots & \vdots \\ \left(a_{j_p} \middle| b_{k_1} \right) & \cdots & \left(a_{j_p} \middle| b_{k_p} \right) \end{vmatrix} I_{\mathcal{F}} \tag{164}$$

An element A of a Clifford Algebra is positive i.e. $A \geq 0$ Iff

$$A = B^{\dagger} B. \tag{165}$$

Thus

$$A \geq 0 \Rightarrow A = A^{\dagger} \tag{166}$$

Note that the square of an operator need not be postive i.e. C^2 could be negative, zero or complex (think of complex eigenvalues)!

1. Outer Nilpotency

One can note that $\wedge$ is the adjoint of $\rfloor$

$$(A \wedge B) * C = A * (B \rfloor C). \tag{167}$$

Thus

$$(A \wedge A) * C = A * (A \rfloor C) \tag{168}$$

and if

$$(A_4 \wedge A_4) = 0 \tag{169}$$

then

$$A_4 * (A_4 \rfloor C) = A_4 * \langle (A_4 \rfloor C) \rangle_4 = 0 \forall C \tag{170}$$

as

$$A_4 \rfloor C = \sum_{0 \le q \le 2r} \langle A_4 C_q \rangle_{q-4} = \langle A_4 C_4 \rangle_0 + \langle A_4 C_5 \rangle_1 + \langle A_4 C_6 \rangle_2 + \langle A_4 C_7 \rangle_3 + \langle A_4 C_8 \rangle_4 + \cdots + \langle A_4 C_{2r} \rangle_{2r-4} \tag{171}$$

hence

$$\langle A_4 \rfloor C \rangle_4 = \langle A_4 C_8 \rangle_4 \tag{172}$$

and

$$A_4 * (A_4 \rfloor C) = A_4 * \langle (A_4 \rfloor C_8) \rangle_4 = 0 \forall C_8. \tag{173}$$

This is equivalent to

$$A_4 * \langle A_4 \rfloor x_{k_1} \wedge \cdots \wedge x_{k_8} \rangle_4 = 0,\ 1 \le k_1 < \cdots < k_8 \le 2r. \tag{174}$$

$$A_4 \rfloor x_{k_1} \wedge \cdots \wedge x_{k_8} = \langle A_4 x_{k_1} \wedge \cdots \wedge x_{k_8} \rangle_4 \tag{175}$$

$$= \sum_{1 \le j_1 < \cdots < j_4 \le 2r} A_{j_1 j_2 j_3 j_4} \langle x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{k_1} x_{k_2} x_{k_3} x_{k_4} x_{k_5} x_{k_6} x_{k_7} x_{k_8} \rangle_4 \tag{176}$$

$$= A_{k_1 k_2 k_3 k_4} x_{k_5} x_{k_6} x_{k_7} x_{k_8} - A_{k_1 k_2 k_3 k_8} x_{k_4} x_{k_5} x_{k_6} x_{k_7} + \cdots \tag{177}$$

$$A_4 * \langle A_4 \rfloor x_{k_1} \wedge \cdots \wedge x_{k_8} \rangle_4 = A_{k_5 k_6 k_7 k_8} A_{k_1 k_2 k_3 k_4} - A_{k_4 k_5 k_6 k_7} A_{k_1 k_2 k_3 k_8} + \cdots = 0 \tag{178}$$

I.e.

$$A_{5678} A_{1234} - A_{4567} A_{1238} + \cdots = 0 \tag{179}$$

Let

$$A_4 = A_{1234} e_{1234} + A_{1235} e_{1235} + A_{4678} e_{4678} + A_{5678} e_{5678}$$

$$A_4 \wedge A_4 \tag{180}$$

$$= (A_{1234} e_{1234} + A_{1235} e_{1235} + A_{4678} e_{4678} + A_{5678} e_{5678}) \tag{181}$$

$$\wedge (A_{1234} e_{1234} + A_{1235} e_{1235} + A_{4678} e_{4678} + A_{5678} e_{5678}) \tag{182}$$

$$= (A_{1234} A_{5678} - A_{1235} A_{4678})\, e_{12345678} \tag{183}$$

$$1234, 1235, 1236, 1237, 1238, \tag{184}$$

$$1245, 1246, 1247, 1248, \tag{185}$$

$$1345, 1346, 1348, \tag{186}$$

$$1256, 1257, 1258, \tag{187}$$

$$1267, 1268, \tag{188}$$

$$1278 \tag{189}$$

$$\binom{4}{1} + \binom{4}{2} + \binom{4}{3} = 4 + 6 + 4 = 14 \tag{190}$$

J. Determinant

The determinant of $F, \det(F) \in \mathcal{F}$ can be defined by

$$\det(F)\mathbf{I} = F\left(\mathbf{I}\right) \tag{191}$$

or equivalently

$$\det(F) = F\left(\mathbf{I}\right)^{\perp} \tag{192}$$

K. Dual (Hodge)

$$A^{\perp} = A\rfloor\mathbf{I}^{-1} = A\mathbf{I}^{-1} \tag{193}$$

where

$$\mathbf{I}^{-1} = \frac{\mathbf{I}^{\dagger}}{\left|\mathbf{I}\right|^{2}} \tag{194}$$

In particular

$$\left(v_1 \wedge \cdots \wedge v_p\right)^{\perp} = v_{p+1} \wedge \cdots \wedge v_{2r} \tag{195}$$

L. Projections and Rejections

A vector a can be decomposed as

$$a = amm^{-1} = \left\{\left(a\left|m\right.\right) + a \wedge m\right\} m^{-1} = a_{\parallel m} + a_{\perp m}, \tag{196}$$

where $a_{\|m}$ is the *Projection* of a onto m and $a_{\perp m}$ is the *Rejection* of a onto m. This can be expressed as

$$a = \mathcal{P}_m(a) + \mathcal{P}_m^{\perp}(a), \tag{197}$$

where

$$\mathcal{P}_m(a) = \left(a \rfloor m^{-1}\right) \rfloor m = |m)\,(m^r|\,a \tag{198}$$

$$\mathcal{P}_m^{\perp}(a) = a - \mathcal{P}_m(a). \tag{199}$$

These definitions generalize to

$$A = \mathcal{P}_B(A) + \mathcal{P}_B^{\perp}(A) \tag{200}$$

$$\mathcal{P}_B(A) = \left(A \rfloor B^{-1}\right) \rfloor A = |B)\,(B^r|\,A) \tag{201}$$

$$\mathcal{P}_B^{\perp}(A) = A - \mathcal{P}_B(A), \tag{202}$$

where A is a general element and B is a blade. E.g.

$$\mathcal{P}_{x_1}(a) = |x_1)\,(x_1^r|\,a) \tag{203}$$

$$\mathcal{P}_{x_1\wedge\cdots\wedge x_p}(a) = \sum_{\substack{1\le p_1<\cdots p_q\le q \\ 1\le q\le p}} \left|x_{p_1}\wedge\cdots\wedge x_{p_q}\right)\left(\left(x_{p_1}\wedge\cdots\wedge x_{p_q}\right)^r\right|a\right) \tag{204}$$

$$= \sum_{\substack{1\le p_1<\cdots p_q\le p \\ 1\le q\le p}} \left|x_{p_1}\wedge\cdots\wedge x_{p_q}\right)\left(\left(x_{p_1}\wedge\cdots\wedge x_{p_q}\right)^r\right|a\right) \tag{205}$$

$$= \sum_{1\le p_1\le p} |x_{p_1})\left(\left(x_{p_1}\right)^r\right|a\right) \tag{206}$$

III. STATES

A. Definition

A state is a positive normalized linear functional

$$f : \mathcal{F} \to \mathbb{C} \tag{207}$$

$$f(I_{\mathcal{F}}) = 1 \tag{208}$$

$$f = f^{\dagger} \tag{209}$$

$$f : \mathcal{F}_{sa} \to \mathbb{R}. \tag{210}$$

It is extremal if it cannot be decomposed into a convex sum of two other states. A non extremal state can be expressed as a convex sum of extremal states i.e.

$$f = \sum_{1 \leq j \leq p} \lambda_j f_j. \tag{211}$$

B. Density Operators

A state f is a Normal state if there exists a density operator $D = D^{\dagger} \in \mathcal{F}$ such that

$$f(A) = \operatorname{tr}\left\{A^{\dagger} D\right\} \; \forall A \in \mathcal{F} \tag{212}$$

$$A^{\dagger} * D = \sum_{0 \leq j \leq 2^{2r}} \left\langle A_j^{\dagger} D_j \right\rangle_0 = \sum_{0 \leq j \leq 2^{2r}} \operatorname{tr}\left\{\left\langle A_j^{\dagger} D_j \right\rangle_0\right\} I_{\mathcal{F}} \tag{213}$$

thus

$$f(A) = \sum_{0 \leq j \leq 2^{2r}} \operatorname{tr}\left\{\left\langle A_j^{\dagger} D_j \right\rangle_0\right\}. \tag{214}$$

In particular as

$$D = \sum_{0 \leq j \leq 2^{2r}} \langle D \rangle_j \equiv \sum_{0 \leq j \leq 2^{2r}} D_j \tag{215}$$

one has

$$\langle I_{\mathcal{F}} D \rangle_0 = D_0 \tag{216}$$

and

$$f_D(I_{\mathcal{F}}) = \operatorname{tr}\{D_0\} = 1, \tag{217}$$

thus

$$D = I_{\mathcal{F}} + \sum_{1 \leq j \leq 2^{4r}} D_j. \tag{218}$$

If $\{p_j\}$ is a set of primitive self adjoint idempotents of $\mathcal{F}$ then

$$I_{\mathcal{F}} = \sum_{1 \leq j \leq 2^{2r}} p_j \tag{219}$$

and

$$f(I_{\mathcal{F}}) = 1 = \sum_{1 \leq j \leq p} f(p_j) = \sum_{1 \leq j,k \leq p} \lambda_k f_k(p_j) \Leftrightarrow f_k(p_j) = \delta_{jk} \tag{220}$$

and thus there is a $1-1$ correspondence between extremal states and primitive idempotents of $\mathcal{F}$.

C. Extension Condition

One has the relation

$$I_{\mathcal{F}} \geq p \geq 0 \tag{221}$$

for all self adjoint idempotents p.

As $f\left(I_{\mathcal{F}}\right)=1$ one has

$$f\left(I_{\mathcal{F}}\right)=\sum_{1\leq j\leq q} f\left(p_j\right)=1, \tag{222}$$

for any resolution of the identity

$$\sum_{1\leq j\leq q} p_j = I_{\mathcal{F}}, \tag{223}$$

thus

$$0 \leq f\left(p_j\right) \leq 1; 1 \leq j \leq q. \tag{224}$$

The necessary and sufficient condition for f to be a state is

$$0 \leq f\left(p\right) \leq 1 \quad \& \quad f\left(I_{\mathcal{F}}\right)=1 \tag{225}$$

for a generating set, $\mathfrak{P}$, *of self adjoint idempotents* p. A generating set is the minimal set of projectors needed to generate all projectors in a subspace of $\mathcal{F}$

$$ \tag{226}$$

Necessary conditions for a positive extension to exist are

$$0 \leq f\left(p\right) \leq 1 \tag{227}$$

for self adjoint idempotents $p \in \mathfrak{P}$ and

$$f\left(I_{\mathcal{F}}\right)=1, \tag{228}$$

sufficient conditions are

$$0 \leq f\left(p\right) \leq 1 \forall p \in P \quad \& \quad f\left(I_{\mathcal{F}}\right)=1 \tag{229}$$

for any set of idempotents containing a generating set .

D. Expectation Values and States

The standard Hamiltonian of molecular and atomic systems is of the form

$$H = \eta_0 I_{\mathcal{F}} + i \sum_{1 \le j_1 < j_2 \le 2r} \eta_{2 j_1 j_2} x_{j_1} x_{j_2} + \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} \eta_{4 j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4} \tag{230}$$

and the energy in a state f_D is given by

$$\begin{aligned} f_D(H) &= \operatorname{tr}\left\{\langle H_0 D_0 \rangle_0\right\} + \operatorname{tr}\left\{\langle H_2 D_2 \rangle_0\right\} + \operatorname{tr}\left\{\langle H_4 D_4 \rangle_0\right\} = \operatorname{tr}\left\{H * D\right\} \\ &= \operatorname{tr}\left\{H_0 * D_0 + H_2 * D_2 + H_4 * D_4\right\} \\ &= \eta_0 + \sum_{1 \le j_1 < j_2 \le 2r} \eta_{2 j_1 j_2} d_{2 j_2 j_1} + \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} \eta_{4 j_1 j_2 j_3 j_4} d_{4 j_1 j_2 j_3 j_4}. \end{aligned} \tag{231}$$

The variables $\{d_{2 j_2 j_1}\}$ and $\{d_{4 j_1 j_2 j_3 j_4}\}$ are constained by the positivity requirement.

If the state D is pure then

$$D^2 = D, \tag{232}$$

while if it is mixed (ensemble) then

$$D \geq D^2 \tag{233}$$

$$\Updownarrow \tag{234}$$

$$D = \sum_{1 \le j \le q} \alpha_j D_j, \tag{235}$$

where $\{D_j\}$ are pure density operators and

$$\sum_{1 \le j \le q} \alpha_j = 1; \qquad 0 \le \alpha_j \le 1. \tag{236}$$

Some properties

$$D^2 = \sum_{0 \le p,q \le 2r} \sum_{j=0}^{\min(p,q)} \langle D_p D_q \rangle_{|p-q|+2j} = \sum_{0 \le n \le 2r} \left\langle D^2 \right\rangle_n, \tag{237}$$

In general

$$\left\langle D^2 \right\rangle_n \neq \langle D \rangle_n. \tag{238}$$

The condition for a state to be a pure state is

$$\sum_{0 \le p,q \le 2r} \sum_{j=0}^{\min(p,q)} \langle D_p D_q \rangle_{|p-q|+2j} = \sum_{0 \le p \le 2r} D_p. \tag{239}$$

e.g. $r = 3$ and

$$D = D_0 + D_2 + D_4 + D_6 \tag{240}$$

then

$$\begin{aligned}
&\sum_{0\le p,q\le 6} \sum_{j=0}^{\min(p,q)} \left\langle D_p D_q \right\rangle_{|p-q|+2j} \\
&= \left\langle D_0 D_0 \right\rangle_0 + \left\langle D_2 D_2 \right\rangle_0 + \left\langle D_2 D_2 \right\rangle_2 + \left\langle D_2 D_2 \right\rangle_4 + \left\langle D_4 D_4 \right\rangle_0 + \left\langle D_4 D_4 \right\rangle_2 + \left\langle D_4 D_4 \right\rangle_4 + \left\langle D_4 D_4 \right\rangle_6 + \left\langle D_4 D_4 \right\rangle_8 \\
&+ \left\langle D_6 D_6 \right\rangle_0 + \left\langle D_6 D_6 \right\rangle_2 + \left\langle D_6 D_6 \right\rangle_4 + \left\langle D_6 D_6 \right\rangle_6 + \left\langle D_6 D_6 \right\rangle_8 + \left\langle D_6 D_6 \right\rangle_{10} + \left\langle D_6 D_6 \right\rangle_{12} \\
&+ 2 \left\{ \begin{array}{c} \left\langle D_0 D_0 \right\rangle_0 + \left\langle D_0 D_2 \right\rangle_0 + \left\langle D_0 D_2 \right\rangle_2 + \left\langle D_0 D_4 \right\rangle_0 + \left\langle D_0 D_4 \right\rangle_2 + \left\langle D_0 D_4 \right\rangle_4 + \left\langle D_0 D_6 \right\rangle_0 \\ + \left\langle D_0 D_6 \right\rangle_2 + \left\langle D_0 D_6 \right\rangle_4 + \left\langle D_0 D_6 \right\rangle_6 \end{array} \right. \\
&+ \left\langle D_2 D_4 \right\rangle_0 + \left\langle D_2 D_4 \right\rangle_2 + \left\langle D_2 D_4 \right\rangle_4 + \left\langle D_2 D_4 \right\rangle_6 + \left\langle D_2 D_6 \right\rangle_0 + \left\langle D_2 D_6 \right\rangle_2 + \left\langle D_2 D_6 \right\rangle_4 \\
&+ \left\langle D_2 D_6 \right\rangle_6 + \left\langle D_2 D_6 \right\rangle_8 && (241) \\
&+ \left\langle D_4 D_6 \right\rangle_0 + \left\langle D_4 D_6 \right\rangle_2 + \left\langle D_4 D_6 \right\rangle_4 + \left\langle D_4 D_6 \right\rangle_6 + \left\langle D_4 D_6 \right\rangle_8 + \left\langle D_4 D_6 \right\rangle_{10} \} && (242)
\end{aligned}$$

then

$$\begin{aligned}
D_0 &= \left\langle D_0 D_0 \right\rangle_0 + \left\langle D_2 D_2 \right\rangle_0 + \left\langle D_4 D_4 \right\rangle_0 + \left\langle D_6 D_6 \right\rangle_0 + 2 \left\{ \begin{array}{c} \left\langle D_0 D_2 \right\rangle_0 + \left\langle D_0 D_4 \right\rangle_0 + \left\langle D_0 D_6 \right\rangle_0 \\ + \left\langle D_2 D_4 \right\rangle_0 + \left\langle D_2 D_6 \right\rangle_0 + \left\langle D_4 D_6 \right\rangle_0 \end{array} \right\} \\
D_2 &= \left\langle D_2 D_2 \right\rangle_2 + \left\langle D_4 D_4 \right\rangle_2 + \left\langle D_6 D_2 \right\rangle_4 + 2 \left\{ \begin{array}{c} \left\langle D_0 D_2 \right\rangle_2 + \left\langle D_0 D_4 \right\rangle_2 + \left\langle D_0 D_6 \right\rangle_2 + \left\langle D_2 D_4 \right\rangle_2 \\ + \left\langle D_2 D_6 \right\rangle_2 + \left\langle D_4 D_6 \right\rangle_2 \end{array} \right\} \\
D_4 &= \left\langle D_2 D_2 \right\rangle_4 + \left\langle D_4 D_4 \right\rangle_4 + \left\langle D_6 D_6 \right\rangle_4 + 2 \left\{ \begin{array}{c} \left\langle D_0 D_4 \right\rangle_4 + \left\langle D_0 D_6 \right\rangle_4 + \left\langle D_2 D_4 \right\rangle_4 \\ + \left\langle D_2 D_6 \right\rangle_4 + \left\langle D_4 D_6 \right\rangle_4 \end{array} \right\} \\
0 &= \left\langle D_4 D_4 \right\rangle_6 + \left\langle D_6 D_6 \right\rangle_6 + \left\langle D_0 D_6 \right\rangle_6 + \left\langle D_2 D_4 \right\rangle_6 + \left\langle D_2 D_6 \right\rangle_6 + \left\langle D_4 D_6 \right\rangle_6 \\
0 &= \left\langle D_4 D_4 \right\rangle_8 + \left\langle D_6 D_6 \right\rangle_8 + \left\langle D_2 D_6 \right\rangle_8 + \left\langle D_4 D_6 \right\rangle_8 \\
0 &= \left\langle D_6 D_6 \right\rangle_{10} + \left\langle D_4 D_6 \right\rangle_{10} \\
0 &= \left\langle D_6 D_6 \right\rangle_{12} . && (243)
\end{aligned}$$

Noting that

$$D_0 = I_{\mathcal{F}} \tag{244}$$

one obtains that (noting $\langle A_j B_k \rangle_0 = A_j * B_k = \langle A_j B_k \rangle_0 \delta_{jk}$)

$$0 = \langle D_2 D_2 \rangle_0 + \langle D_4 D_4 \rangle_0 + \langle D_6 D_6 \rangle_0 \tag{245}$$

$$D_2 = -\left\{ \langle D_2 D_2 \rangle_2 + \langle D_4 D_4 \rangle_2 + \langle D_6 D_2 \rangle_4 + 2\left\{ \langle D_2 D_4 \rangle_2 + \langle D_2 D_6 \rangle_2 + \langle D_4 D_6 \rangle_2 \right\} \right\} \tag{246}$$

$$D_4 = -\left\{ \langle D_2 D_2 \rangle_4 + \langle D_4 D_4 \rangle_4 + \langle D_6 D_6 \rangle_4 + 2\left\{ \langle D_2 D_4 \rangle_4 + \langle D_2 D_6 \rangle_4 + \langle D_4 D_6 \rangle_4 \right\} \right\} \tag{247}$$

$$0 = \langle D_4 D_4 \rangle_6 + \langle D_6 D_6 \rangle_6 + \langle D_6 \rangle_6 + \langle D_2 D_4 \rangle_6 + \langle D_2 D_6 \rangle_6 + \langle D_4 D_6 \rangle_6 \tag{248}$$

$$0 = \langle D_4 D_4 \rangle_8 + \langle D_6 D_6 \rangle_8 + \langle D_2 D_6 \rangle_8 + \langle D_4 D_6 \rangle_8 \tag{249}$$

$$0 = \langle D_6 D_6 \rangle_{10} + \langle D_4 D_6 \rangle_{10} \tag{250}$$

$$0 = \langle D_6 D_6 \rangle_{12} . \tag{251}$$

The norms are given by

$$|D|^2 = \langle DD \rangle_0 = \left\langle D^2 \right\rangle_0 = \sum_{0 \le p \le 2r} D_p * D_p = \sum_{0 \le p \le 2r} \left\langle D_p^2 \right\rangle = \sum_{0 \le p \le 2r} |D_p|^2 \tag{252}$$

$$\left|D^2\right|^2 = \left\langle D^2 D^2 \right\rangle_0 = \left\langle D^4 \right\rangle_0 = \sum_{0 \le p \le 2r} D_p^2 * D_p^2 = \sum_{0 \le p \le 2r} \left\langle D_p^4 \right\rangle_0 = \sum_{0 \le p \le 2r} \left|D_p^2\right|^2 . \tag{253}$$

1. *The Number Operator*

If a state describes on the average N particles then

$$f_D(\mathcal{N}) = \operatorname{tr}\left\{ \langle \mathcal{N}_0 D_0 \rangle_0 \right\} + \operatorname{tr}\left\{ \langle \mathcal{N}_2 D_2 \rangle_0 \right\} = N, \tag{254}$$

where the number operator $\mathcal{N}$ is

$$\mathcal{N} = \sum_{1 \le j \le r} n_j = \mathcal{N}_0 + \mathcal{N}_2 = \mathcal{N} = \frac{r}{2} I_{\mathcal{F}} + i \sum_{1 \le j \le r} x_j x_{j+r} . \tag{255}$$

If it describes N particles then

$$[\mathcal{N}, D] = 0 \,\&\, \operatorname{tr}\left\{ \langle \mathcal{N} D \rangle_0 \right\} = N$$
$$\Updownarrow$$
$$\operatorname{tr}\left\{ \left\langle \mathcal{N}^2 D \right\rangle_0 \right\} - \left(\operatorname{tr}\left\{ \langle \mathcal{N} D \rangle_0 \right\} \right)^2 = 0, \tag{256}$$

i.e.

$$f_D(\mathcal{N})^2 - f_D\left(\mathcal{N}^2\right) = 0. \tag{257}$$

The number operator expressed in the $\{x_j\}$ bases involves r, which is infinity when $r=\infty$, however the back transformation works as one gets $\infty-\infty$ in the form of $r-r$ as can be seen from

$$\begin{aligned}
\mathcal{N} &= \frac{r}{2}I_{\mathcal{F}}+i\sum_{1\le j\le r}\frac{1}{\sqrt{2}}\left(a_j^\dagger+a_j\right)\frac{i}{\sqrt{2}}\left(a_j^\dagger-a_j\right)\\
&= \frac{r}{2}I_{\mathcal{F}}+i\sum_{1\le j\le r}\frac{i}{2}\left(-a_j^\dagger a_j+a_ja_j^\dagger\right)\\
&= \frac{r}{2}I_{\mathcal{F}}-\frac{1}{2}\sum_{1\le j\le r}\left(-a_j^\dagger a_j+I_{\mathcal{F}}-a_j^\dagger a_j\right)\\
&= \frac{r}{2}I_{\mathcal{F}}-\sum_{1\le j\le r}\left(\frac{1}{2}I_{\mathcal{F}}-a_j^\dagger a_j\right)\\
&= \frac{r}{2}I_{\mathcal{F}}-\frac{r}{2}I_{\mathcal{F}}+\sum_{1\le j\le r}a_j^\dagger a_j\\
&= \mathcal{N}.
\end{aligned}\tag{258}$$

The square of the number operator is given by

$$\mathcal{N}^2=\sum_{1\le j,k\le r}n_jn_k=\sum_{1\le j\le r}n_j+2\sum_{1\le j<k\le r}n_jn_k=\sum_{1\le j\le r}n_j+2\sum_{1\le j<k\le r}a_j^\dagger a_ja_k^\dagger a_k \tag{259}$$

$$=\sum_{1\le j\le r}n_j+2\sum_{1\le j<k\le r}a_j^\dagger a_k^\dagger a_ka_j=\sum_{1\le j\le r}n_j+2\sum_{1\le j<k\le r}n_{jk}=\mathcal{N}+2\mathcal{N}_2, \tag{260}$$

which gives in the Clifford basis

$$\begin{aligned}
\mathcal{N}^2 &= \left(\frac{r}{2}I_{\mathcal{F}}+i\sum_{1\le j\le r}x_jx_{j+r}\right)\left(\frac{r}{2}I_{\mathcal{F}}+i\sum_{1\le k\le r}x_kx_{k+r}\right)\\
&= \frac{r^2}{4}I_{\mathcal{F}}+i\frac{r}{2}\sum_{1\le j\le r}x_jx_{j+r}+\sum_{1\le j,k\le r}x_jx_kx_{j+r}x_{k+r}\\
&= \left(\frac{r^2}{4}+\frac{r}{4}\right)I_{\mathcal{F}}+i\frac{r}{2}\sum_{1\le j\le r}x_jx_{j+r}+2\sum_{1\le j<k\le r}x_jx_kx_{j+r}x_{k+r}\\
&= \frac{r\left(r+1\right)}{4}I_{\mathcal{F}}+i\frac{r}{2}\sum_{1\le j\le r}x_jx_{j+r}+2\sum_{1\le j<k\le r}x_jx_kx_{j+r}x_{k+r}.
\end{aligned}\tag{261}$$

Thus a state describes a fixed number of particles if

$$f_D\left(\mathcal{N}\right)^2-f_D\left(\mathcal{N}^2\right)=f_D\left(\mathcal{N}\right)^2-f_D\left(\mathcal{N}\right)-2f_D\left(\mathcal{N}_2\right)=f_D\left(\mathcal{N}\right)\left(f_D\left(\mathcal{N}\right)-1\right)-2f_D\left(\mathcal{N}_2\right)=0, \tag{262}$$

i.e.

$$f_D\left(\mathcal{N}\right)\left(f_D\left(\mathcal{N}\right)-1\right)=2f_D\left(\mathcal{N}_2\right). \tag{263}$$

The pair operator $\mathcal{N}_2$ can be expanded as

$$\mathcal{N}_2 = \sum_{1\le j<k\le r} n_j n_k = \sum_{1\le j<k\le r} \left(\frac{1}{2} I_{\mathcal{F}} + i x_j x_{j+r}\right)\left(\frac{1}{2} I_{\mathcal{F}} + i x_k x_{k+r}\right) \tag{264}$$

$$= \frac{1}{4}\binom{r}{2} I_{\mathcal{F}} + 2i \sum_{1\le j\le r} x_j x_{j+r} - \sum_{1\le j<k\le r} x_j x_{j+r} x_k x_{k+r} = \langle \mathcal{N}_2\rangle_0 + \langle \mathcal{N}_2\rangle_2 + \langle \mathcal{N}_2\rangle_4 . \tag{265}$$

IV. IDEMPOTENTS

If an idempotent p is self adjoint

$$p = p^{\dagger} \tag{266}$$

it is positive as

$$p = p^2 = p^{\dagger} p. \tag{267}$$

It is possible to have non self adjoint idempotents (theses are associated with non orthogonal projectors and nonorthogonal subspace decompositions), which are non positive.

A. Primitive Idempotents

A primitive idempotent could be of any order i.e.

$$p = \sum_{0\le j\le 2^{2r}} \langle p\rangle_j = \sum_{0\le j\le 2^{2r}} p_j, \tag{268}$$

where $p_j = \langle p\rangle_j$ and is not in general idempotent. The scalar part can be expressed as

$$p * I_{\mathcal{F}} = p_0 = \operatorname{tr}\left\{p^{\dagger} I_{\mathcal{F}}\right\} I_{\mathcal{F}}. \tag{269}$$

As $p = p^2$ one has in particular that

$$p_0 = \left\langle p^2\right\rangle_0 = \left\langle p^{\dagger} p\right\rangle_0 \ge 0 = \sum_{0\le j\le 2^{2r}} \left\langle p_j^{\dagger} p_j\right\rangle_0 = p_0^2 + \sum_{1\le j\le 2^{2r}} \left\langle p_j^{\dagger} p_j\right\rangle_0 \tag{270}$$

$$p_0 - p_0^2 = p_0\left(I_{\mathcal{F}} - p_0\right) = \sum_{1\le j\le 2^{2r}} \left\langle p_j^{\dagger} p_j\right\rangle_0 \ge 0, \tag{271}$$

so

$$0 \le p_0 = \operatorname{tr}\left\{p_0 I_{\mathcal{F}}\right\} I_{\mathcal{F}} \le I_{\mathcal{F}} \Longleftrightarrow 0 \le \operatorname{tr}\left\{p_0 I_{\mathcal{F}}\right\} \le 1. \tag{272}$$

B. Non Primitive Idempotents

1. Simple Idempotents

a. Order 1 (Elementary)

$$n_j = a_j^\dagger a_j = \sum_{\substack{1\le p\le r \\ 1\le j_2<\cdots<j_p\le r}} \left|\varphi_j\varphi_{j_2}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_j\varphi_{j_2}\cdots\varphi_{j_p}\right| = \frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r} \tag{273}$$

$$h_j = a_ja_j^\dagger = \frac{1}{2}I_{\mathcal{F}} - n_j = \frac{1}{2}\sum_{\substack{0\le p\le r \\ 1\le j_1<\cdots<j_p\le r}} \left|\varphi_{j_1}\varphi_{j_2}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_{j_1}\varphi_{j_2}\cdots\varphi_{j_p}\right| \tag{274}$$

$$- \sum_{\substack{1\le p\le r \\ 1\le j_2<\cdots<j_p\le r}} \left|\varphi_j\varphi_{j_2}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_j\varphi_{j_2}\cdots\varphi_{j_p}\right| \tag{275}$$

$$= \frac{1}{2}I_{\mathcal{F}} - ix_jx_{j+r}. \tag{276}$$

b. Order 2 (Non-Elementary)

$$a_j^\dagger a_k^\dagger a_ka_j = a_j^\dagger a_ja_k^\dagger a_k = n_jn_k = \sum_{3\le p\le r} \left|\varphi_j\varphi_k\varphi_{j_2}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_j\varphi_k\varphi_{j_2}\cdots\varphi_{j_p}\right| \tag{277}$$

$$= \left(\frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} + ix_kx_{k+r}\right) \tag{278}$$

$$n_jh_k = \left(\frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} - ix_kx_{k+r}\right) \tag{279}$$

$$h_jh_k = \left(\frac{1}{2}I_{\mathcal{F}} - ix_jx_{j+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} - ix_kx_{k+r}\right) \tag{280}$$

In general simple (Non-Elementary) non primitive idempotents are products of simple idempotents e.g. an idempotent of order $n+m$ is given by

$$p_{n,m} = \prod_{l=1,\cdots,m} n_i \prod_{l=m+1,\cdots,n} h_i = \prod_{l=1,\cdots,m}\left(\frac{1}{2}I_{\mathcal{F}} + ix_lx_{l+r}\right)\prod_{l=m+1,\cdots,n}\left(\frac{1}{2}I_{\mathcal{F}} - ix_lx_{l+r}\right) \tag{281}$$

so that e.g.

$$p_{2,0} = \left(\frac{1}{2}I_{\mathcal{F}} + ix_1x_{1+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} + ix_2x_{2+r}\right) = \frac{1}{4}I_{\mathcal{F}} + ix_1x_{1+r} + ix_2x_{2+r} + x_1x_2x_{1+r}x_{2+r} \tag{282}$$

$$p_{1,1} = \left(\frac{1}{2}I_{\mathcal{F}} + ix_1x_{1+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} - ix_2x_{2+r}\right) = \frac{1}{4}I_{\mathcal{F}} + ix_1x_{1+r} - ix_2x_{2+r} - x_1x_2x_{1+r}x_{2+r} \tag{283}$$

$$p_{0,2} = \left(\frac{1}{2}I_{\mathcal{F}} - ix_1x_{1+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} - ix_2x_{2+r}\right) = \frac{1}{4}I_{\mathcal{F}} - ix_1x_{1+r} - ix_2x_{2+r} + x_1x_2x_{1+r}x_{2+r}. \tag{284}$$

One can also see (below) that Primitive idempotents are also products of Elementary idempotents.

2. Non Simple Idempotents

3. Construction of Idempotents

Let $\{y_j|\, 1 \leq j \leq 2r\}$ be an orthonormal basis of $\mathcal{F}_1$ that generates the basis $\left\{I_{\mathcal{F}}, \left\{y_{j_1}\cdots y_{j_p}\middle|\, 1 \leq j_1 < \cdots < j_p \leq 2r; 1 \leq p \leq 2r\right\}\right\}$ of $\mathcal{F}$.Let

$$p_{+j_1} = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right) \tag{285}$$

$$p_{-j_1} = \frac{1}{2}\left(I_{\mathcal{F}} - y_{j_1}\right) \tag{286}$$

then

$$p_{+j_1}^2 = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right)\frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right) = \frac{1}{4}\left(I_{\mathcal{F}} + 2y_{j_1} + y_{j_1}^2\right) = \frac{1}{4}\left(2I_{\mathcal{F}} + 2y_{j_1}\right) = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right) = p_{+j_1} \tag{287}$$

$$p_{-j_1}^2 = \frac{1}{2}\left(I_{\mathcal{F}} - y_{j_1}\right)\frac{1}{2}\left(I_{\mathcal{F}} - y_{j_1}\right) = \frac{1}{4}\left(I_{\mathcal{F}} - 2y_{j_1} + y_{j_1}^2\right) = \frac{1}{4}\left(2I_{\mathcal{F}} - 2y_{j_1}\right) = \frac{1}{2}\left(I_{\mathcal{F}} - y_{j_1}\right) = p_{-j_1} \tag{288}$$

$$p_{+j_1}p_{-j_1} = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right)\frac{1}{2}\left(I_{\mathcal{F}} - y_{j_1}\right) = \frac{1}{4}\left(I_{\mathcal{F}} - y_{j_1}^2\right) = 0 \tag{289}$$

$$p_{-j_1} + p_{+j_1} = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right) + \frac{1}{2}\left(I_{\mathcal{F}} - y_{j_1}\right) = I_{\mathcal{F}} \tag{290}$$

and

$$p_{+j_1}p_{+j_2} = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right)\frac{1}{2}\left(I_{\mathcal{F}} + y_{j_2}\right) = \frac{1}{4}\left(I_{\mathcal{F}} + y_{j_1} + y_{j_2} + y_{j_1}y_{j_2}\right) \neq p_{+j_2}p_{+j_1} \tag{291}$$

$$p_{+j_2}p_{+j_1} = \frac{1}{2}\left(I_{\mathcal{F}} + y_{j_2}\right)\frac{1}{2}\left(I_{\mathcal{F}} + y_{j_1}\right) = \frac{1}{4}\left(I_{\mathcal{F}} + y_{j_1} + y_{j_2} - y_{j_1}y_{j_2}\right) \neq p_{+j_1}p_{+j_2}. \tag{292}$$

In general if

$$A^2 = I_{\mathcal{F}} \tag{293}$$

then one can form the idempotents

$$A_+ = \frac{1}{2}\left(I_{\mathcal{F}} + A\right) \tag{294}$$

$$A_- = \frac{1}{2}\left(I_{\mathcal{F}} - A\right) \tag{295}$$

such that

$$A_+A_- = 0 \tag{296}$$

$$A_+ + A_- = I_{\mathcal{F}}. \tag{297}$$

$\lceil$

$$\frac{1}{2}\left(I_{\mathcal{F}} + A\right)\frac{1}{2}\left(I_{\mathcal{F}} + A\right) = \frac{1}{4}\left(I_{\mathcal{F}} + 2A + I_{\mathcal{F}}\right) = \frac{1}{2}\left(I_{\mathcal{F}} + A\right) \tag{298}$$

$$\frac{1}{2}\left(I_{\mathcal{F}} - A\right)\frac{1}{2}\left(I_{\mathcal{F}} - A\right) = \frac{1}{4}\left(I_{\mathcal{F}} - 2A + I_{\mathcal{F}}\right) = \frac{1}{2}\left(I_{\mathcal{F}} - A\right) \tag{299}$$

$$\frac{1}{2}\left(I_{\mathcal{F}} + A\right)\frac{1}{2}\left(I_{\mathcal{F}} - A\right) = \frac{1}{4}\left(I_{\mathcal{F}} - I_{\mathcal{F}}\right) = 0 \tag{300}$$

$\rfloor$

The $\{n_j\}$ and $\{h_j\}$ are such pairs of projectors.

The inverse relationship is

$$A = 2A_+ - I_{\mathcal{F}} = 2A_- + I_{\mathcal{F}}. \tag{301}$$

If

$$A = A^\dagger \tag{302}$$

then the idempotents are self adjoint.

C. Ket-Bra Primitive Idempotents

Recall

$$n_j = a_j^\dagger a_j = \frac{1}{2} I_{\mathcal{F}} + i x_j x_{j+r} \tag{303}$$

$$h_j = a_j a_j^\dagger = I_{\mathcal{F}} - n_j = \frac{1}{2} I_{\mathcal{F}} - i x_j x_{j+r}. \tag{304}$$

Let $r = 3$

$$\mathcal{H}_{\mathcal{F}} = \text{linspan}\left\{ |\phi\rangle, |\varphi_1\rangle, |\varphi_2\rangle, |\varphi_3\rangle, |\varphi_1\varphi_2\rangle, |\varphi_1\varphi_3\rangle, |\varphi_2\varphi_3\rangle, |\varphi_1\varphi_2\varphi_3\rangle \right\} \tag{305}$$

then 1.

$$|\phi\rangle\langle\phi| = a_3 a_2 a_1 a_1^\dagger a_2^\dagger a_3^\dagger = a_3 a_3^\dagger a_2 a_2^\dagger a_1 a_1^\dagger = h_3 h_2 h_1 = \prod_{l=1,2,3} h_i \tag{306}$$

as

2.

$$|\varphi_1\rangle\langle\varphi_1| = a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger = a_3 a_3^\dagger a_2 a_2^\dagger a_1^\dagger a_1 = h_3 h_2 n_1 = \prod_{l=1} n_i \prod_{l=2,3} h_i \tag{307}$$

as

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\phi\rangle = 0 \tag{308}$$

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\varphi_1\rangle = |\varphi_1\rangle \tag{309}$$

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\varphi_2\rangle = 0 \tag{310}$$

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\varphi_3\rangle = 0 \tag{311}$$

$$a_3 a_2 a_1^\dagger a_1 a_3^\dagger a_2^\dagger |\varphi_1\varphi_2\rangle = 0 \tag{312}$$

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\varphi_1\varphi_3\rangle = 0 \tag{313}$$

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\varphi_2\varphi_3\rangle = 0 \tag{314}$$

$$a_3 a_2 a_1^\dagger a_1 a_2^\dagger a_3^\dagger |\varphi_1\varphi_2\varphi_3\rangle = 0 \tag{315}$$

3.

$$|\varphi_1\varphi_2\rangle\langle\varphi_1\varphi_2| = a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger = a_3 a_3^\dagger a_1^\dagger a_1 a_2^\dagger a_2 = h_3 n_2 n_1 = \prod_{l=1,2} n_i \prod_{l=3} h_i \tag{316}$$

as

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\phi\right\rangle = 0 \tag{317}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_1\right\rangle = 0 \tag{318}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_2\right\rangle = 0 \tag{319}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_3\right\rangle = 0 \tag{320}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_1\varphi_2\right\rangle = \left|\varphi_1\varphi_2\right\rangle \tag{321}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_1\varphi_3\right\rangle = 0 \tag{322}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_2\varphi_3\right\rangle = 0 \tag{323}$$

$$a_3 a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger \left|\varphi_1\varphi_2\varphi_3\right\rangle = 0 \tag{324}$$

4.

$$\left|\varphi_1\varphi_2\varphi_3\right\rangle \left\langle\varphi_1\varphi_2\varphi_3\right| = a_1^\dagger a_1 a_2^\dagger a_2 a_3^\dagger a_3 = n_3 n_2 n_1 = \prod_{l=1,2,3} n_i \tag{325}$$

Thus in general

$$\left|\varphi_{j_1}\cdots\varphi_{j_p}\right\rangle \left\langle\varphi_{j_1}\cdots\varphi_{j_p}\right| = \prod_{0\le l\le p} n_i \prod_{p+1\le l\le r} h_i = \prod_{0\le l\le p} n_i \prod_{p+1\le l\le r} (I_{\mathcal{F}} - n_l) \tag{326}$$

$$= \prod_{0\le l\le p} \left\{\frac{1}{2} I_{\mathcal{F}} + i x_l x_{l+r}\right\} \prod_{p+1\le l\le r} \left\{\frac{1}{2} I_{\mathcal{F}} - i x_l x_{l+r}\right\}. \tag{327}$$

1. Transition Operators

Let $r = 5$ then

$$\left|\varphi_2\right\rangle \left\langle\varphi_1\right| = a_5 a_5^\dagger a_4 a_4^\dagger a_3 a_3^\dagger a_2^\dagger a_1 \tag{328}$$

$$\left|\varphi_2\varphi_3\right\rangle \left\langle\varphi_1\varphi_2\right| = a_5 a_5^\dagger a_4 a_4^\dagger a_2^\dagger a_3^\dagger a_2 a_1 \tag{329}$$

$$\left|\varphi_3\varphi_4\right\rangle \left\langle\varphi_1\varphi_2\right| = a_5 a_5^\dagger a_3^\dagger a_4^\dagger a_2 a_1 \tag{330}$$

$$\left|\varphi_3\varphi_4\varphi_4\right\rangle \left\langle\varphi_1\varphi_2\varphi_3\right| = a_5 a_5^\dagger a_2^\dagger a_3^\dagger a_4^\dagger a_3 a_2 a_1 \tag{331}$$

They give the correct products as

$$\left|\varphi_2\right\rangle \left\langle\varphi_1\right| \left|\varphi_3\varphi_4\right\rangle \left\langle\varphi_1\varphi_2\right| = a_5 a_5^\dagger a_4 a_4^\dagger a_3 a_3^\dagger a_2^\dagger a_1 a_5 a_5^\dagger a_3^\dagger a_4^\dagger a_2 a_1 = 0 \tag{332}$$

$$|\varphi_3\varphi_4\rangle\langle\varphi_1\varphi_2|\,|\varphi_3\varphi_4\rangle\langle\varphi_1\varphi_2| = a_5a_5^\dagger a_3^\dagger a_4^\dagger a_2a_1a_5a_5^\dagger a_3^\dagger a_4^\dagger a_2a_1 = 0 \tag{333}$$

$$|\varphi_1\varphi_2\rangle\langle\varphi_3\varphi_4|\,|\varphi_3\varphi_4\rangle\langle\varphi_1\varphi_2| = a_1^\dagger a_2^\dagger a_4a_3a_5a_5^\dagger a_5a_5^\dagger a_3^\dagger a_4^\dagger a_2a_1 = a_1^\dagger a_2^\dagger a_4a_3a_5a_5^\dagger\left(1-a_5^\dagger a_5\right)a_3^\dagger a_4^\dagger a_2a_1 \tag{334}$$

$$= a_1^\dagger a_1a_2^\dagger a_2a_5a_5^\dagger a_3a_3^\dagger a_4a_4^\dagger = |\varphi_1\varphi_2\rangle\langle\varphi_1\varphi_2|\,. \tag{335}$$

so in general

$$|\phi\rangle\langle\phi| = \prod_{1\le j\le r} h_j \tag{336}$$

$$|\varphi_{j_1}\rangle = \left\{\prod_{1\le j\ne j_1\le r} h_j\right\} a_{j_1}^\dagger \tag{337}$$

$$\langle\varphi_{j_1}| = \left\{\prod_{1\le j\ne j_1\le r} h_j\right\} a_{j_1}. \tag{338}$$

$$|\varphi_{j_2}\rangle\langle\varphi_{j_1}| = \prod_{1\le j\ne j_1,j_2\le rj} h_ja_{j_2}^\dagger a_{j_1} \tag{339}$$

$$|\varphi_{j_3}\varphi_{j_4}\rangle\langle\varphi_{j_1}\varphi_{j_2}| = \prod_{1\le j\ne j_1,j_2,j_3,j_4\le r} h_ja_{j_3}^\dagger a_{j_4}^\dagger a_{j_2}a_{j_1} \tag{340}$$

$$|\varphi_{j_4}\varphi_{j_5}\varphi_{j_6}\rangle\langle\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}| = \prod_{1\le j\ne j_1,j_2,j_3,j_4,j_5,j_6\le r} h_ja_{j_4}^\dagger a_{j_5}^\dagger a_{j_6}^\dagger a_{j_3}a_{j_2}a_{j_1}, \tag{341}$$

$$\left|\varphi_{j_1}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_q}\right| = \prod_{1\le j\ne j_1,\cdots,,j_q,,k_1,\cdots,k_p\le r} h_ja_{j_1}^\dagger\cdots a_{j_q}^\dagger a_{k_p}\cdots a_{k_1} \tag{342}$$

which gives the idempotents as well.

The general form of the products can be seen from

$$|\varphi_{j_2}\rangle\langle\varphi_{j_1}|\,|\varphi_{j_3}\varphi_{j_4}\rangle\langle\varphi_{j_1}\varphi_{j_2}| = \prod_{1\le j\ne j_1,j_2\le r} h_ja_{j_2}^\dagger a_{j_1}\prod_{1\le j\ne j_3,j_4,j_5,j_6\le r} h_ja_{j_5}^\dagger a_{j_6}^\dagger a_{j_4}a_{j_3} = 0. \tag{343}$$

this gives

$$\langle\varphi_{j_1}|\,|\varphi_{j_1}\rangle = \left\{\prod_{1\le j\ne j_1\le r} h_j\right\} a_{j_1}\left\{\prod_{1\le j\ne j_1\le r} h_j\right\} a_{j_1}^\dagger = \prod_{1\le j\le r} h_j = 1 \tag{344}$$

$$|\varphi_{j_1}\rangle\langle\varphi_{j_1}| = \left\{\prod_{1\le j\ne j_1\le r} h_j\right\} a_{j_1}^\dagger\left\{\prod_{1\le j\ne j_1\le r} h_j\right\} a_{j_1} = n_1\prod_{2\le l\le r} h_i \tag{345}$$

2. Idempotents formed from non blades

The following idempotent is primitive

$$\left|g\right\rangle\left\langle g\right| = \sum_{1\leq j,k\leq s} c_j c_k \left|\varphi_j \varphi_{j+s}\right\rangle\left\langle \varphi_k \varphi_{k+s}\right| = \sum_{1\leq j,k\leq s}\left\{ c_j c_k a_j^{\dagger} a_{j+s}^{\dagger} a_k a_{k+s} \prod_{1\leq l\neq j,j+s,k,k+s\leq r} h_l \right\}. \tag{346}$$

and not a product of idempotents.

In general

$$\left|\varphi_{j_1}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_q}\right| = \prod_{1\leq j\neq j_1,\cdots,,j_q,,k_1,\cdots,k_p\leq r} h_j a_{j_1}^{\dagger}\cdots a_{j_q}^{\dagger} a_{k_p}\cdots a_{k_1} = \tag{347}$$

$$\prod_{1\leq j\neq j_1,\cdots,,j_q,,k_1,\cdots,k_p\leq r}\left\{\frac{1}{2}I_{\mathcal{F}} - i x_j x_{j+r}\right\}\frac{1}{\sqrt{2}}\left(x_{j_1} - i x_{j_1+r}\right)\cdots\frac{1}{\sqrt{2}}\left(x_{j_q} - i x_{j_q+r}\right)\frac{1}{\sqrt{2}}\left(x_{k_p} + i x_{k_p+r}\right) \tag{348}$$

$$\cdots\frac{1}{\sqrt{2}}\left(x_{k_1} + i x_{k_1+r}\right), \tag{349}$$

thus *the most general primitive idempotent* is

$$\left|\Psi\right\rangle\left\langle\Psi\right| = \sum_{0\leq p,q\leq r}\sum_{\substack{1\leq j_1<\cdots<j_q\leq r\\ 1\leq k_1<\cdots<k_p\leq r}} \bar{c}_{k_1\cdots k_p} c_{j_1\cdots j_q}\left|\varphi_{j_1}\cdots\varphi_{j_p}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_q}\right| \tag{350}$$

$$= \sum_{0\leq p,q\leq r}\sum_{\substack{1\leq j_1<\cdots<j_q\leq r\\ 1\leq k_1<\cdots<k_p\leq r}} \bar{c}_{k_1\cdots k_p} c_{j_1\cdots j_q} \tag{351}$$

$$\times\prod_{1\leq j\neq j_1,\cdots,,j_q,,k_1,\cdots,k_p\leq r}\left\{\frac{1}{2}I_{\mathcal{F}} - i x_j x_{j+r}\right\}\frac{1}{\sqrt{2}}\left(x_{j_1} - i x_{j_1+r}\right)\cdots\frac{1}{\sqrt{2}}\left(x_{j_q} - i x_{j_q+r}\right) \tag{352}$$

$$\times\frac{1}{\sqrt{2}}\left(x_{k_p} + i x_{k_p+r}\right)\cdots\frac{1}{\sqrt{2}}\left(x_{k_1} + i x_{k_1+r}\right). \tag{353}$$

D. Self Adjoint Idempotents in $\mathcal{F}_{(4)} = \mathcal{F}_0 \oplus \mathcal{F}_2 \oplus \mathcal{F}_4$

The following set of commuting nonprimitive selfadjoint idempotents

$$\left\{I_{\mathcal{F}}, n_1, \cdots, n_r, n_1 n_2, \cdots, n_{r-1} n_r\right\} \tag{354}$$

are contained in $\mathcal{F}_{(4)}$, the condition that they generate further idempotents in $\mathcal{F}_{(4)}$ is

$$\left(\alpha I_{\mathcal{F}} + \sum_{1\leq j\leq r}\alpha_j n_j + \sum_{1\leq j<k\leq r}\alpha_{jk} n_j n_k\right)^2 = \alpha I_{\mathcal{F}} + \sum_{1\leq j\leq r}\alpha_j n_j + \sum_{1\leq j<k\leq r}\alpha_{jk} n_j n_k. \tag{355}$$

The LHS term gives

$$\left(\alpha I_{\mathcal{F}} + \sum_{1\le j\le r} \alpha_{1j} n_j + \sum_{1\le j<k\le r} \alpha_{2jk} n_j n_k\right)^2 = \left(\alpha I_{\mathcal{F}} + \sum_{1\le j\le r} \alpha_{1j} n_j\right)^2$$

$$+ 2\left(\alpha I_{\mathcal{F}} + \sum_{1\le j\le r} \alpha_{1j} n_j\right) \sum_{1\le j<k\le r} \alpha_{2jk} n_j n_k + \left(\sum_{1\le j<k\le r} \alpha_{2jk} n_j n_k\right)^2$$

$$= \alpha I_{\mathcal{F}} + \sum_{1\le j\le r} \left\{4\alpha\alpha_{1j} + \alpha_{1j}^2\right\} n_j + 2 \sum_{1\le j<k\le r} \alpha_{1j}\alpha_{1k} n_j n_k + \sum_{1\le j,k,l\le r} \alpha_{1j}\alpha_{2kl} n_j n_k n_l$$

$$+ \frac{1}{4} \sum_{1\le j,k,l,m\le r} \alpha_{2jk}\alpha_{2lm} n_j n_k n_l n_m, \tag{356}$$

noting that

$$\sum_{1\le j,k,l\le r} \alpha_{1j}\alpha_{2kl} n_j n_k n_l = \frac{1}{6} \sum_{1\le j<k<l\le r} \alpha_{1j}\alpha_{2kl} n_j n_k n_l + \sum_{1\le j,l\le r} \alpha_{1j}\alpha_{2jl} n_j n_j n_l + \sum_{1\le j,k\le r} \alpha_{1j}\alpha_{2kj} n_j n_k n_j$$

$$= \frac{1}{6} \sum_{1\le j<k<l\le r} \alpha_{1j}\alpha_{2kl} n_j n_k n_l + 2 \sum_{1\le j,l\le r} \alpha_{1j}\alpha_{2jl} n_j n_l \tag{357}$$

and

$$\frac{1}{4} \sum_{1\le j,k,l,m\le r} \alpha_{2jk}\alpha_{2lm} n_j n_k n_l n_m = \sum_{1\le j<k<l<m\le r} \left(\alpha_{2jk}\alpha_{2lm} + \alpha_{2jl}\alpha_{2km} + \alpha_{2jm}\alpha_{2lk}\right) n_j n_k n_l n_m$$

$$+ \frac{1}{4} \sum_{1\le j,k,m\le r} \alpha_{2jk}\alpha_{2jm} n_j n_k n_m$$

$$+ \frac{1}{4} \sum_{1\le j,k,l\le r} \alpha_{2jk}\alpha_{2lj} n_j n_k n_l + \frac{1}{4} \sum_{1\le j,k,m\le r} \alpha_{2jk}\alpha_{2km} n_j n_k n_m + \frac{1}{4} \sum_{1\le j,k,l\le r} \alpha_{2jk}\alpha_{2lk} n_j n_k n_l$$

$$+ \sum_{1\le j,k,l,m\le r} \alpha_{2jk}\alpha_{2jk} n_j n_k \tag{358}$$

leading to

$$\left(\alpha_0 I_{\mathcal{F}} + \sum_{1\le j\le r} \alpha_{1j} n_j + \sum_{1\le j<k\le r} \alpha_{2jk} n_j n_k\right)^2 = \alpha_0^2 I_{\mathcal{F}} + \sum_{1\le j\le r} \left\{4\alpha_0\alpha_{1j} + \alpha_{1j}^2\right\} n_j$$

$$+ 2 \sum_{1\le j<k\le r} \alpha_{1j}\alpha_{1k} n_j n_k + 2 \sum_{1\le j,l\le r} \alpha_{1j}\alpha_{2jl} n_j n_l + \frac{1}{6} \sum_{1\le j<k<l\le r} \alpha_{1j}\alpha_{2kl} n_j n_k n_l$$

$$+ \sum_{1\le j<k<l<m\le r} \left(\alpha_{2jk}\alpha_{2lm} + \alpha_{2jl}\alpha_{2km} + \alpha_{2jm}\alpha_{2lk}\right) n_j n_k n_l n_m \tag{359}$$

⌈Let $r = 4$ then

$$
\begin{aligned}
&(\alpha_0 I_{\mathcal{F}} + \alpha_1 n_1 + \alpha_2 n_2 + \alpha_3 n_3 + \alpha_4 n_4 + \alpha_{12} n_1 n_2 \\
&+ \alpha_{13} n_1 n_3 + \alpha_{14} n_1 n_4 + \alpha_{23} n_2 n_3 + \alpha_{24} n_2 n_4 + \alpha_{34} n_3 n_4)^2 \\
&= (\alpha_0 I_{\mathcal{F}} + \alpha_1 n_1 + \alpha_2 n_2 + \alpha_3 n_3 + \alpha_4 n_4)^2 \\
&+ 2\left(\alpha_0 I_{\mathcal{F}} + \alpha_1 n_1 + \alpha_2 n_2 + \alpha_3 n_3 + \alpha_4 n_4\right)\left(\alpha_{12} n_1 n_2 + \alpha_{13} n_1 n_3 + \alpha_{14} n_1 n_4 + \alpha_{23} n_2 n_3 + \alpha_{24} n_2 n_4 + \alpha_{34} n_3 n_4\right) \\
&+ \left(\alpha_{12} n_1 n_2 + \alpha_{13} n_1 n_3 + \alpha_{14} n_1 n_4 + \alpha_{23} n_2 n_3 + \alpha_{24} n_2 n_4 + \alpha_{34} n_3 n_4\right)^2
\end{aligned}
\tag{360}
$$

The first term gives

$$
\begin{aligned}
&(\alpha_0 I_{\mathcal{F}} + \alpha_1 n_1 + \alpha_2 n_2 + \alpha_3 n_3 + \alpha_4 n_4)^2 \\
&= \alpha_0^2 I_{\mathcal{F}} + \alpha_1^2 n_1 + \alpha_2^2 n_2 + \alpha_3^2 n_3 + \alpha_4^2 n_4 \\
&+ 2\left(\alpha_1 \alpha_2 n_1 n_2 + \alpha_1 \alpha_3 n_1 n_3 + \alpha_1 \alpha_4 n_1 n_4 + \alpha_2 \alpha_3 n_2 n_3 + \alpha_2 \alpha_4 n_2 n_4 + \alpha_3 \alpha_4 n_3 n_4\right)
\end{aligned}
\tag{361}
$$

The second term gives

$$
\begin{aligned}
&2\left(\alpha_0 I_{\mathcal{F}} + \alpha_1 n_1 + \alpha_2 n_2 + \alpha_3 n_3 + \alpha_4 n_4\right)\left(\alpha_{12} n_1 n_2 + \alpha_{13} n_1 n_3 + \alpha_{14} n_1 n_4 + \alpha_{23} n_2 n_3 + \alpha_{24} n_2 n_4 + \alpha_{34} n_3 n_4\right) \\
&= 2\alpha_0 \alpha_{12} n_1 n_2 + 2\alpha_0 \alpha_{13} n_1 n_3 + 2\alpha_0 \alpha_{14} n_1 n_4 + 2\alpha_0 \alpha_{23} n_2 n_3 + 2\alpha_0 \alpha_{24} n_2 n_4 + 2\alpha_0 \alpha_{34} n_3 n_4 \\
&+ 2\alpha_1 \alpha_{12} n_1 n_2 + 2\alpha_{13} \alpha_1 n_1 n_3 + 2\alpha_{14} \alpha_1 n_1 n_4 + 2\alpha_{23} \alpha_1 n_1 n_2 n_3 + 2\alpha_{24} \alpha_1 n_1 n_2 n_4 + 2\alpha_{34} \alpha_1 n_1 n_3 n_4 \\
&+ 2\alpha_{12} \alpha_2 n_1 n_2 + 2\alpha_{13} \alpha_2 n_1 n_2 n_3 + 2\alpha_{14} \alpha_2 n_1 n_2 n_4 + 2\alpha_{23} \alpha_2 n_2 n_3 + 2\alpha_{24} \alpha_2 n_2 n_4 + 2\alpha_{34} \alpha_2 n_2 n_3 n_4 \\
&+ 2\alpha_{13} \alpha_2 n_1 n_2 n_3 + 2\alpha_{13} \alpha_3 n_1 n_3 + 2\alpha_{14} \alpha_3 n_1 n_3 n_4 + 2\alpha_{23} \alpha_3 n_2 n_3 + 2\alpha_{24} \alpha_3 n_2 n_3 n_4 + 2\alpha_{34} \alpha_3 n_3 n_4 \\
&+ 2\alpha_4 \alpha_{12} n_1 n_2 n_4 + 2\alpha_{13} \alpha_4 n_1 n_3 n_4 + 2\alpha_{14} \alpha_4 n_1 n_4 + 2\alpha_{23} \alpha_4 n_2 n_3 n_4 + 2\alpha_{24} \alpha_4 n_2 n_4 + 2\alpha_{34} \alpha_4 n_3 n_4 \\
&= 2\left(\alpha_0 \alpha_{12} + \alpha_1 \alpha_{12} + \alpha_2 \alpha_{12}\right) n_1 n_2 + 2\left(\alpha_0 \alpha_{13} + \alpha_1 \alpha_{13} + \alpha_{31} \alpha_{13}\right) n_1 n_3 + 2\left(\alpha_0 \alpha_{14} + \alpha_1 \alpha_{14} + \alpha_4 \alpha_{14}\right) n_1 n_4 \\
&+ 2\left(\alpha_0 \alpha_{23} + \alpha_2 \alpha_{23} + \alpha_3 \alpha_{23}\right) n_2 n_3 + 2\left(\alpha_0 \alpha_{24} + \alpha_2 \alpha_{24} + \alpha_4 \alpha_{24}\right) n_2 n_4 + 2\left(\alpha_0 \alpha_{34} + \alpha_3 \alpha_{34} + \alpha_4 \alpha_{34}\right) n_3 n_4 \\
&+ 2\left(\alpha_1 \alpha_{23} + \alpha_2 \alpha_{13} + \alpha_3 \alpha_{12}\right) n_1 n_2 n_3 + 2\left(\alpha_1 \alpha_{24} + \alpha_2 \alpha_{14} + \alpha_4 \alpha_{12}\right) n_1 n_2 n_4 \\
&+ 2\left(\alpha_1 \alpha_{34} + \alpha_3 \alpha_{14} + \alpha_4 \alpha_{13}\right) n_1 n_3 n_4 + 2\left(\alpha_2 \alpha_{34} + \alpha_3 \alpha_{24} + \alpha_4 \alpha_{23}\right) n_2 n_3 n_4
\end{aligned}
\tag{362}
$$

and the third term

$$
\begin{aligned}
&\left(\alpha_{12} n_1 n_2 + \alpha_{13} n_1 n_3 + \alpha_{14} n_1 n_4 + \alpha_{23} n_2 n_3 + \alpha_{24} n_2 n_4 + \alpha_{34} n_3 n_4\right) \\
&\times \left(\alpha_{12} n_1 n_2 + \alpha_{13} n_1 n_3 + \alpha_{14} n_1 n_4 + \alpha_{23} n_2 n_3 + \alpha_{24} n_2 n_4 + \alpha_{34} n_3 n_4\right) \\
&= \alpha_{12}^2 n_1 n_2 + \alpha_{13}^2 n_1 n_3 + \alpha_{14}^2 n_1 n_4 + \alpha_{23}^2 n_2 n_3 + \alpha_{24}^2 n_2 n_4 + \alpha_{34}^2 n_3 n_4 \\
&+ 2\alpha_{12}\alpha_{13} n_1 n_2 n_3 + 2\alpha_{12}\alpha_{14} n_1 n_2 n_4 + 2\alpha_{12}\alpha_{23} n_1 n_2 n_3 + 2\alpha_{12}\alpha_{24} n_1 n_2 n_4 + 2\alpha_{12}\alpha_{34} n_1 n_2 n_3 n_4 \\
&+ 2\alpha_{13}\alpha_{14} n_1 n_3 n_4 + 2\alpha_{13}\alpha_{23} n_1 n_2 n_3 + 2\alpha_{13}\alpha_{24} n_1 n_2 n_4 + 2\alpha_{13}\alpha_{34} n_1 n_3 n_4 \\
&+ 2\alpha_{14}\alpha_{23} n_1 n_2 n_3 n_4 + 2\alpha_{14}\alpha_{24} n_1 n_2 n_4 + 2\alpha_{14}\alpha_{34} n_1 n_3 n_4 \\
&+ 2\alpha_{23}\alpha_{24} n_2 n_3 n_4 + 2\alpha_{23}\alpha_{34} n_2 n_3 n_4 \\
&+ 2\alpha_{24}\alpha_{34} n_2 n_3 n_4 \\
&= \alpha_{12}^2 n_1 n_2 + \alpha_{13}^2 n_1 n_3 + \alpha_{14}^2 n_1 n_4 + \alpha_{23}^2 n_2 n_3 + \alpha_{24}^2 n_2 n_4 + \alpha_{34}^2 n_3 n_4 \\
&+ 2\left(\alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23}\right) n_1 n_2 n_3 + 2\left(\alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24}\right) n_1 n_2 n_4 \\
&+ 2\left(\alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34}\right) n_1 n_3 n_4 + 2\left(\alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34}\right) n_2 n_3 n_4 \\
&+ 2\left(\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34}\right) n_1 n_2 n_3 n_4
\end{aligned}
\tag{363}
$$

Adding the terms together one obtains

$$
\begin{aligned}
&\alpha_0^2 I_{\mathcal{F}} + \alpha_1^2 n_1 + \alpha_2^2 n_2 + \alpha_3^2 n_3 + \alpha_4^2 n_4 \\
&+ 2\left(\left(\alpha_1\alpha_2 + \alpha_0\alpha_{12} + \alpha_1\alpha_{12} + \alpha_2\alpha_{12} + \frac{1}{2}\alpha_{12}^2\right) n_1 n_2\right. \\
&+ \left(\alpha_1\alpha_3 + \alpha_0\alpha_{13} + \alpha_1\alpha_{13} + \alpha_3\alpha_{13} + \frac{1}{2}\alpha_{13}^2\right) n_1 n_3 \\
&+ \left(\alpha_1\alpha_4 + \alpha_0\alpha_{14} + \alpha_1\alpha_{14} + \alpha_4\alpha_{14} + \frac{1}{2}\alpha_{14}^2\right) n_1 n_4 \\
&+ \left(\alpha_2\alpha_3 + \alpha_0\alpha_{23} + \alpha_2\alpha_{23} + \alpha_3\alpha_{23} + \frac{1}{2}\alpha_{23}^2\right) n_2 n_3 \\
&+ \left(\alpha_2\alpha_4 + \alpha_0\alpha_{24} + \alpha_2\alpha_{24} + \alpha_4\alpha_{24} + \frac{1}{2}\alpha_{24}^2\right) n_2 n_4 \\
&+ \left(\alpha_3\alpha_4 + \alpha_0\alpha_{34} + \alpha_3\alpha_{34} + \alpha_4\alpha_{34} + \frac{1}{2}\alpha_{34}^2\right) n_3 n_4 \\
&+ \left(\alpha_1\alpha_{23} + \alpha_2\alpha_{13} + \alpha_3\alpha_{12} + \alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23}\right) n_1 n_2 n_3 \\
&+ \left(\alpha_1\alpha_{24} + \alpha_2\alpha_{14} + \alpha_4\alpha_{12} + \alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24}\right) n_1 n_2 n_4 \\
&+ \left(\alpha_1\alpha_{34} + \alpha_3\alpha_{14} + \alpha_4\alpha_{13} + \alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34}\right) n_1 n_3 n_4 \\
&+ \left(\alpha_2\alpha_{34} + \alpha_3\alpha_{24} + \alpha_4\alpha_{23} + \alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34}\right) n_2 n_3 n_4 \\
&+ \left.\left(\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34}\right) n_1 n_2 n_3 n_4\right). \qquad (364)
\end{aligned}
$$

Thus the idempotency condition gives

$$
\begin{aligned}
&\alpha_0 \left(\alpha_0 - 1\right) I_{\mathcal{F}} + \alpha_1 \left(\alpha_1 - 1\right) n_1 + \alpha_2 \left(\alpha_2 - 1\right) n_2 + \alpha_3 \left(\alpha_3 - 1\right) n_3 + \alpha_4 \left(\alpha_4 - 1\right) n_4 \\
&+ \left(2\alpha_1\alpha_2 + 2\alpha_0\alpha_{12} + 2\alpha_1\alpha_{12} + 2\alpha_2\alpha_{12} + \alpha_{12}^2 - \alpha_{12}\right) n_1 n_2 \\
&+ \left(2\alpha_1\alpha_3 + \alpha_0\alpha_{13} + 2\alpha_1\alpha_{13} + 2\alpha_3\alpha_{13} + \alpha_{13}^2 - \alpha_{13}\right) n_1 n_3 \\
&+ \left(2\alpha_1\alpha_4 + 2\alpha_0\alpha_{14} + \alpha_1\alpha_{14} + 2\alpha_4\alpha_{14} + \alpha_{14}^2 - \alpha_{14}\right) n_1 n_4 \\
&+ \left(2\alpha_2\alpha_3 + 2\alpha_0\alpha_{23} + \alpha_2\alpha_{23} + 2\alpha_3\alpha_{23} + \alpha_{23}^2 - \alpha_{23}\right) n_2 n_3 \\
&+ \left(2\alpha_2\alpha_4 + 2\alpha_0\alpha_{24} + \alpha_2\alpha_{24} + 2\alpha_4\alpha_{24} + \alpha_{24}^2 - \alpha_{24}\right) n_2 n_4 \\
&+ \left(2\alpha_3\alpha_4 + 2\alpha_0\alpha_{34} + \alpha_3\alpha_{34} + 2\alpha_4\alpha_{34} + \alpha_{34}^2 - \alpha_{24}\right) n_3 n_4 \\
&+ 2\left(\alpha_1\alpha_{23} + \alpha_2\alpha_{13} + \alpha_3\alpha_{12} + \alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23}\right) n_1 n_2 n_3 \\
&+ 2\left(\alpha_1\alpha_{24} + \alpha_2\alpha_{14} + \alpha_4\alpha_{12} + \alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24}\right) n_1 n_2 n_4 \\
&+ 2\left(\alpha_1\alpha_{34} + \alpha_3\alpha_{14} + \alpha_4\alpha_{13} + \alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34}\right) n_1 n_3 n_4 \\
&+ 2\left(\alpha_2\alpha_{34} + \alpha_3\alpha_{24} + \alpha_4\alpha_{23} + \alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34}\right) n_2 n_3 n_4 \\
&+ 2\left(\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34}\right) n_1 n_2 n_3 n_4 = 0.
\end{aligned}
\tag{365}
$$

Thus we obtain the simultaneous equations

$$\alpha_0 \left(\alpha_0 - 1\right) = 0 \Leftrightarrow \alpha_0 = 0 \text{ or } 1 \tag{366}$$

$$\alpha_1 \left(\alpha_1 - 1\right) = 0 \Leftrightarrow \alpha_1 = 0 \text{ or } 1 \tag{367}$$

$$\alpha_2 \left(\alpha_2 - 1\right) = 0 \Leftrightarrow \alpha_2 = 0 \text{ or } 1 \tag{368}$$

$$\alpha_3 \left(\alpha_3 - 1\right) = 0 \Leftrightarrow \alpha_3 = 0 \text{ or } 1 \tag{369}$$

$$\alpha_4 \left(\alpha_4 - 1\right) = 0 \Leftrightarrow \alpha_4 = 0 \text{ or } 1 \tag{370}$$

$$
\begin{aligned}
&2\alpha_1\alpha_2 + 2\alpha_0\alpha_{12} + 2\alpha_1\alpha_{12} + 2\alpha_2\alpha_{12} + \alpha_{12}^2 - \alpha_{12} = 0\\
&2\alpha_1\alpha_3 + 2\alpha_0\alpha_{13} + 2\alpha_1\alpha_{13} + 2\alpha_3\alpha_{13} + \alpha_{13}^2 - \alpha_{13} = 0\\
&2\alpha_1\alpha_4 + 2\alpha_0\alpha_{14} + 2\alpha_1\alpha_{14} + 2\alpha_4\alpha_{14} + \alpha_{14}^2 - \alpha_{14} = 0\\
&2\alpha_2\alpha_3 + 2\alpha_0\alpha_{23} + 2\alpha_2\alpha_{23} + 2\alpha_3\alpha_{23} + \alpha_{23}^2 - \alpha_{23} = 0\\
&2\alpha_2\alpha_4 + 2\alpha_0\alpha_{24} + 2\alpha_2\alpha_{24} + 2\alpha_4\alpha_{24} + \alpha_{24}^2 - \alpha_{24} = 0\\
&2\alpha_3\alpha_4 + 2\alpha_0\alpha_{34} + 2\alpha_3\alpha_{34} + 2\alpha_4\alpha_{34} + \alpha_{34}^2 - \alpha_{34} = 0\\
&\alpha_1\alpha_{23} + \alpha_2\alpha_{13} + \alpha_3\alpha_{12} + \alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23} = 0\\
&\alpha_1\alpha_{24} + \alpha_2\alpha_{14} + \alpha_4\alpha_{12} + \alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24} = 0\\
&\alpha_1\alpha_{34} + \alpha_3\alpha_{14} + \alpha_4\alpha_{13} + \alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34} = 0\\
&\alpha_2\alpha_{34} + \alpha_3\alpha_{24} + \alpha_4\alpha_{23} + \alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34} = 0\\
&\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34} = 0. \qquad (371)
\end{aligned}
$$

1. Solutions

1. The first solution is trivial $\alpha_0 = 1, \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_{12} = \alpha_{13} = \alpha_{14} = \alpha_{23} = \alpha_{24} = \alpha_{34} = 0$.

2. Let $\alpha_0 = \alpha_2 = \alpha_3 = \alpha_4 = 0$, $\alpha_1 = 1$ then

$$
\begin{aligned}
&\alpha_{12} + \alpha_{12}^2 = 0 \Leftrightarrow \alpha_{12}\left(\alpha_{12} + 1\right) = 0 \Leftrightarrow \alpha_{12} = -1 \text{ or } 0\\
&\alpha_{13} + \alpha_{13}^2 = 0 \Leftrightarrow \alpha_{13}\left(\alpha_{13} + 1\right) = 0 \Leftrightarrow \alpha_{13} = -1 \text{ or } 0\\
&\alpha_{14} + \alpha_{14}^2 = 0 \Leftrightarrow \alpha_{14}\left(\alpha_{14} + 1\right) = 0 \Leftrightarrow \alpha_{14} = -1 \text{ or } 0\\
&\alpha_{23}^2 = 0 \Leftrightarrow \alpha_{23} = 0\\
&\alpha_{24}^2 = 0 \Leftrightarrow \alpha_{24} = 0\\
&\alpha_{34}^2 = 0 \Leftrightarrow \alpha_{34} = 0\\
&\alpha_{12}\alpha_{13} = 0\\
&\alpha_{12}\alpha_{14} = 0\\
&\alpha_{13}\alpha_{14} = 0.
\end{aligned}
$$

Thus the solutions are

$$n_1 \tag{372}$$

$$n_1 - n_1 n_2 \tag{373}$$

$$n_1 - n_1 n_3 \tag{374}$$

$$n_1 - n_1 n_4. \tag{375}$$

We get analogous solutions when $\alpha_j = 1; 2 \leq j \leq 4$ and everything else is zero.

3. Let $\alpha_0 = 0, \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$ then we obtain the simultaneous equations

$$\begin{aligned}
&\alpha_{12}^2 - \alpha_{12} = 0 \Leftrightarrow \alpha_{12} = -1 \text{ or } 0 \\
&\alpha_{13}^2 - \alpha_{13} = 0 \Leftrightarrow \alpha_{13} = -1 \text{ or } 0 \\
&\alpha_{14}^2 - \alpha_{14} = 0 \Leftrightarrow \alpha_{14} = -1 \text{ or } 0 \\
&\alpha_{23}^2 - \alpha_{23} = 0 \Leftrightarrow \alpha_{23} = -1 \text{ or } 0 \\
&\alpha_{24}^2 - \alpha_{24} = 0 \Leftrightarrow \alpha_{24} = -1 \text{ or } 0 \\
&\alpha_{34}^2 - \alpha_{34} = 0 \Leftrightarrow \alpha_{34} = -1 \text{ or } 0 \\
&\alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23} = 0 \\
&\alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24} = 0 \\
&\alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34} = 0 \\
&\alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34} = 0 \\
&\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34} = 0.
\end{aligned} \tag{376}$$

Thus the solutions are

$$n_j n_k; 1 \leq j < k \leq 4$$

4. Let $\alpha_0 = 1, \alpha_2 = \alpha_3 = \alpha_4 = 0,\ \alpha_1 = 1$ then we obtain the simultaneous equations

$$\begin{aligned}
&3\alpha_{12} + \alpha_{12}^2 = 0 \Leftrightarrow \alpha_{12}\,(3 + \alpha_{12}) = 0 \Leftrightarrow \alpha_{12} = -3 \text{ or } 0\\
&3\alpha_{13} + \alpha_{13}^2 = 0 \Leftrightarrow \alpha_{13}\,(3 + \alpha_{13}) = 0 \Leftrightarrow \alpha_{13} = -3 \text{ or } 0\\
&3\alpha_{14} + \alpha_{14}^2 = 0 \Leftrightarrow \alpha_{14}\,(3 + \alpha_{14}) = 0 \Leftrightarrow \alpha_{14} = -3 \text{ or } 0\\
&\alpha_{23} + \alpha_{23}^2 = 0 \Leftrightarrow \alpha_{23}\,(1 + \alpha_{23}) = 0 \Leftrightarrow \alpha_{23} = -1 \text{ or } 0\\
&\alpha_{24} + \alpha_{24}^2 = 0 \Leftrightarrow \alpha_{24}\,(1 + \alpha_{24}) = 0 \Leftrightarrow \alpha_{24} = -1 \text{ or } 0\\
&\alpha_{34} + \alpha_{34}^2 = 0 \Leftrightarrow \alpha_{34}\,(1 + \alpha_{34}) = 0 \Leftrightarrow \alpha_{34} = -1 \text{ or } 0\\
&\alpha_{23} + \alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23} = 0\\
&\alpha_{24} + \alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24} = 0\\
&\alpha_{34} + \alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34} = 0\\
&\alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34} = 0\\
&\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34} = 0.
\end{aligned} \tag{377}$$

Thus the solutions are

$$I_{\mathcal{F}} - n_1$$

5. Let $\alpha_0 = 1, \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$ then we obtain the simultaneous equations

$$\begin{aligned}
&\alpha_{12} + \alpha_{12}^2 = 0 \Leftrightarrow \alpha_{12} = -1 \text{ or } 0\\
&\alpha_{13} + \alpha_{13}^2 = 0 \Leftrightarrow \alpha_{13} = -1 \text{ or } 0\\
&\alpha_{14} + \alpha_{14}^2 = 0 \Leftrightarrow \alpha_{14} = -1 \text{ or } 0\\
&\alpha_{23} + \alpha_{23}^2 = 0 \Leftrightarrow \alpha_{23} = -1 \text{ or } 0\\
&\alpha_{24} + \alpha_{24}^2 = 0 \Leftrightarrow \alpha_{24} = -1 \text{ or } 0\\
&\alpha_{34} + \alpha_{34}^2 = 0 \Leftrightarrow \alpha_{34} = -1 \text{ or } 0\\
&\alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23} = 0\\
&\alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24} = 0\\
&\alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34} = 0\\
&\alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34} = 0\\
&\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34} = 0.
\end{aligned} \tag{378}$$

Thus the solutions are

$$I_{\mathcal{F}} - n_j n_k; 1 \le j < k \le 4.$$

6. Let $\alpha_0 = 0, \alpha_1 = \alpha_2 = 1, \alpha_3 = \alpha_4 = 0$ then we obtain the simultaneous equations

$$\begin{aligned}
&2 + 2\alpha_{12} + 2\alpha_{12} + \alpha_{12}^2 - \alpha_{12} = 0 \Leftrightarrow 2 + 3\alpha_{12} + \alpha_{12}^2 = 0 \Leftrightarrow \alpha_{12} = -2 \text{ or } -1 \\
&2\alpha_{13} + \alpha_{13}^2 - \alpha_{13} = 0 \Leftrightarrow \alpha_{13} + \alpha_{13}^2 = 0 \Leftrightarrow \alpha_{13}\left(1 + \alpha_{13}\right) = 0 \Leftrightarrow \alpha_{13} = 0 \text{ or } -1 \\
&2\alpha_{14} + \alpha_{14}^2 - \alpha_{14} = 0 \Leftrightarrow \alpha_{14} + \alpha_{14}^2 = 0 \Leftrightarrow \alpha_{14}\left(1 + \alpha_{14}\right) = 0 \Leftrightarrow \alpha_{14} = 0 \text{ or } -1 \\
&2\alpha_{23} + \alpha_{23}^2 - \alpha_{23} = 0 \Leftrightarrow \alpha_{23} + \alpha_{23}^2 = 0 \Leftrightarrow \alpha_{23}\left(1 + \alpha_{23}\right) = 0 \Leftrightarrow \alpha_{23} = 0 \text{ or } -1 \\
&2\alpha_{24} + \alpha_{24}^2 - \alpha_{24} = 0 \Leftrightarrow \alpha_{24} + \alpha_{24}^2 = 0 \Leftrightarrow \alpha_{24}\left(1 + \alpha_{24}\right) = 0 \Leftrightarrow \alpha_{24} = 0 \text{ or } -1 \\
&\alpha_{34}^2 - \alpha_{34} = 0 \Leftrightarrow \alpha_{34}\left(\alpha_{34} - 1\right) = 0 \Leftrightarrow \alpha_{34} = 0 \text{ or } 1 \\
&\alpha_{23} + \alpha_{13} + \alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23} = 0 \Leftrightarrow \alpha_{23}\left(1 + \alpha_{12} + \alpha_{13}\right) + \alpha_{13}\left(1 + \alpha_{12}\right) = 0 \\
&\alpha_{24} + \alpha_{14} + \alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24} = 0 \\
&\alpha_{34} + \alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34} = 0 \\
&\alpha_{34} + \alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34} = 0 \\
&\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34} = 0. \qquad (379)
\end{aligned}$$

$$\alpha_{12} = -2 \qquad (380)$$

$$\alpha_{13} = \alpha_{14} = \alpha_{23} = \alpha_{24} = \alpha_{34} = 0 \qquad (381)$$

$$\alpha_{12} = -1 \qquad (382)$$

$$\alpha_{13} = \alpha_{14} = \alpha_{23} = \alpha_{24} = \alpha_{34} = 0 \qquad (383)$$

$$\alpha_{12} = -1 \qquad (384)$$

$$\alpha_{13} = -1; \alpha_{14} = \alpha_{23} = \alpha_{24} = \alpha_{34} = 0 \qquad (385)$$

$$\alpha_{12} = -1 \qquad (386)$$

$$\alpha_{13} = \alpha_{14} = \alpha_{23} = \alpha_{24} = 0; \alpha_{34} = 1; \text{This is not consistent with the second and third last equations} \qquad (387)$$

Thus the solutions are

$$n_1 + n_2 - n_1 n_2$$

$$n_1 + n_2 - 2n_1 n_2$$

$$n_1 + n_2 - n_1 n_2 - n_1 n_3$$

$$n_1 + n_2 - n_1 n_2 - n_2 n_3$$

$$n_1 + n_2 - n_1 n_2 - n_1 n_4$$

$$n_1 + n_2 - n_1 n_2 - n_2 n_4$$

$$\begin{aligned}
2\alpha_1\alpha_2 + 2\alpha_0\alpha_{12} + 2\alpha_1\alpha_{12} + 2\alpha_2\alpha_{12} + \alpha_{12}^2 - \alpha_{12} = 0 \\
2\alpha_1\alpha_3 + 2\alpha_0\alpha_{13} + 2\alpha_1\alpha_{13} + 2\alpha_3\alpha_{13} + \alpha_{13}^2 - \alpha_{13} = 0 \\
2\alpha_1\alpha_4 + 2\alpha_0\alpha_{14} + 2\alpha_1\alpha_{14} + 2\alpha_4\alpha_{14} + \alpha_{14}^2 - \alpha_{14} = 0 \\
2\alpha_2\alpha_3 + 2\alpha_0\alpha_{23} + 2\alpha_2\alpha_{23} + 2\alpha_3\alpha_{23} + \alpha_{23}^2 - \alpha_{23} = 0 \\
2\alpha_2\alpha_4 + 2\alpha_0\alpha_{24} + 2\alpha_2\alpha_{24} + 2\alpha_4\alpha_{24} + \alpha_{24}^2 - \alpha_{24} = 0 \\
2\alpha_3\alpha_4 + 2\alpha_0\alpha_{34} + 2\alpha_3\alpha_{34} + 2\alpha_4\alpha_{34} + \alpha_{34}^2 - \alpha_{34} = 0 \\
\alpha_1\alpha_{23} + \alpha_2\alpha_{13} + \alpha_3\alpha_{12} + \alpha_{12}\alpha_{13} + \alpha_{12}\alpha_{23} + \alpha_{13}\alpha_{23} = 0 \\
\alpha_1\alpha_{24} + \alpha_2\alpha_{14} + \alpha_4\alpha_{12} + \alpha_{12}\alpha_{14} + \alpha_{12}\alpha_{24} + \alpha_{14}\alpha_{24} = 0 \\
\alpha_1\alpha_{34} + \alpha_3\alpha_{14} + \alpha_4\alpha_{13} + \alpha_{13}\alpha_{14} + \alpha_{13}\alpha_{34} + \alpha_{14}\alpha_{34} = 0 \\
\alpha_2\alpha_{34} + \alpha_3\alpha_{24} + \alpha_4\alpha_{23} + \alpha_{23}\alpha_{24} + \alpha_{23}\alpha_{34} + \alpha_{24}\alpha_{34} = 0 \\
\alpha_{12}\alpha_{34} + \alpha_{13}\alpha_{24} + \alpha_{14}\alpha_{34} = 0. \qquad (388)
\end{aligned}$$

E. One and Two particle Operators

Such operators can be expressed in first quantization form as

$$\gamma = \gamma_0 I_{\mathcal{F}} + i \sum_{\substack{1\le l_1\le r \\ 1\le m_1\le r}} \gamma_{1l_1m_1} \sum_{0\le M\le r-1} \sum_{0\le j_1<\cdots<j_M\le r} \left|\varphi_{l_1}\varphi_{j_1}\cdots\varphi_{j_M}\right\rangle \left\langle\varphi_{m_1}\varphi_{j_1}\cdots\varphi_{j_M}\right| \\ + \sum_{\substack{1\le l_1<l_2\le r \\ 1\le m_1<m_2\le r}} \gamma_{2l_1l_2m_1m_2} \sum_{0\le M\le r-2} \sum_{0\le j_1<\cdots<j_M\le r} \left|\varphi_{l_1}\varphi_{l_2}\varphi_{j_1}\cdots\varphi_{j_M}\right\rangle \left\langle\varphi_{m_1}\varphi_{m_2}\varphi_{j_1}\cdots\varphi_{j_M}\right|.$$

Note that

$$I_{\mathcal{F}} = \sum_{0\le M\le r} \sum_{0\le j_1<\cdots<j_M\le r} \left|\varphi_{j_1}\cdots\varphi_{j_M}\right\rangle \left\langle\varphi_{j_1}\cdots\varphi_{j_M}\right| \qquad (389)$$

If γ is a self adjoint idempotent i.e. an orthogonal projector then

$$\gamma = \gamma^2 \geq 0. \tag{390}$$

The expectation value of such operators in a pure N-particle state is

$$\langle \Psi_N | \left\{ \gamma_0 I_{\mathcal{F}} + i \sum_{\substack{1\leq l_1 \leq r \\ 1\leq m_1 \leq r}} \gamma_{1 l_1 m_1} \sum_{1\leq j_1 < \cdots < j_N \leq r} \left| \varphi_{l_1} \varphi_{j_1} \cdots \varphi_{j_{N-1}} \right\rangle \left\langle \varphi_{m_1} \varphi_{j_1} \cdots \varphi_{j_{N-1}} \right| \right. \tag{391}$$

$$\left. + \sum_{\substack{1\leq l_1 < l_2 \leq r \\ 1\leq m_1 < m_2 \leq r}} \gamma_{2 l_1 l_2 m_1 m_2} \sum_{1\leq j_1 < \cdots < j_{N-2} \leq r} \left| \varphi_{l_1} \varphi_{l_2} \varphi_{j_1} \cdots \varphi_{j_{N-2}} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \varphi_{j_1} \cdots \varphi_{j_{N-2}} \right| \right\} \Psi_N \rangle . \tag{392}$$

V. IDEMPOTENCY

A. Order 2 self adjoint idempotents

Let $\{x_j\}$ and $\{y_j\}$ be real orthonormal bases and

$$\gamma = \gamma_0 I_{\mathcal{F}} + i \sum_{1\leq j<k\leq 2r} \gamma_{jk} x_j x_k = \gamma_0 I_{\mathcal{F}} + i \sum_{1\leq j \leq r} c_j y_j y_{j+r};\ c_j \geq 0, \tag{393}$$

where

$$(\mathbf{y}) = (\mathbf{x})\, \mathbb{O} \tag{394}$$

$$\mathbb{O} \in SO(2r, \mathbb{R}). \tag{395}$$

Then its square is

$$\gamma^2 = \left(\gamma_0 I_{\mathcal{F}} + i \sum_{1\leq j \leq r} c_j y_j y_{j+r} \right) \left(\gamma_0 I_{\mathcal{F}} + i \sum_{1\leq k \leq r} c_k y_k y_{k+r} \right) \tag{396}$$

$$= \gamma_0^2 I_{\mathcal{F}} + 2i\gamma_0 \sum_{1\leq j \leq r} c_j y_j y_{j+r} + \sum_{1\leq j \leq r} c_j^2 y_j y_{j+r} y_{j+r} y_j + \sum_{1\leq j \neq k, \leq r} c_j c_k y_j y_{j+r} y_{k+r} y_k \tag{397}$$

$$= \gamma_0^2 I_{\mathcal{F}} + \sum_{1\leq j \leq r} c_j^2 I_{\mathcal{F}} + 2i\gamma_0 \sum_{1\leq j \leq r} c_j y_j y_{j+r} + \sum_{1\leq j \neq k, \leq r} c_j c_k y_j y_{j+r} y_{k+r} y_k \tag{398}$$

$$\left(\gamma_0^2 + \sum_{1\leq j \leq r} c_j^2 \right) I_{\mathcal{F}} + 2i\gamma_0 \sum_{1\leq j \leq r} c_j y_j y_{j+r} + \sum_{1\leq j \neq k, \leq r} c_j c_k y_j y_{j+r} y_{k+r} y_k \tag{399}$$

Hence the condition

$$\gamma^2 - \gamma = 0 \tag{400}$$

gives

$$\left(\gamma_0^2 - \gamma_0 + \sum_{1\le j\le r} c_j^2\right) I_{\mathcal{F}} + 2i\left(\gamma_0 - \frac{1}{2}\right) \sum_{1\le j\le r} c_j y_j y_{j+r} + 2 \sum_{1\le j<k,\le r} c_j c_k y_j y_{j+r} y_{k+r} y_k = 0. \tag{401}$$

Thus

$$\gamma_0 = \frac{1}{2} \tag{402}$$

$$c_j c_k = 0;\ 1 \le j < k \le r \Longleftrightarrow c_j = 0; 2 \le j < k \le r \tag{403}$$

$$\sum_{1\le j\le r} c_j^2 = \frac{1}{4} = c_1^2 \Rightarrow c_1 = \pm\frac{1}{2}. \tag{404}$$

All this gives

$$\gamma = \frac{1}{2} I_{\mathcal{F}} + \frac{i}{2} y_1 y_{1+r} = \frac{1}{2}\left(I_{\mathcal{F}} \pm i y_1 y_{1+r}\right). \tag{405}$$

B. Order 4 self adjoint idempotents

The idempotents are of the form

$$\gamma = \gamma_0 I_{\mathcal{F}} + i \sum_{1\le j\le r} c_j y_j y_{j+r} + \sum_{1\le j_1<j_2<j_3<j_4\le 2r} \gamma_{j_1 j_2 j_3 j_4} y_{j_1} y_{j_2} y_{j_3} y_{j_4} \tag{406}$$

$$\begin{aligned}
\gamma^2 &= \gamma_0^2 I_{\mathcal{F}} + \left\langle \gamma_4^2 \right\rangle_0 + \left\langle \gamma_2^2 \right\rangle_0 \\
&+ 2\left\langle \gamma_2 \gamma_4 \right\rangle_2 + 2\gamma_0\gamma_2 \\
&+ \left\langle \gamma_4^2 \right\rangle_4 + \left\langle \gamma_2^2 \right\rangle_4 + 2\gamma_0\gamma_4 \\
&+ 2\left\langle \gamma_2 \gamma_4 \right\rangle_6 + \left\langle \gamma_4^2 \right\rangle_8,
\end{aligned}$$

thus

$$\left\langle \gamma_4^2 \right\rangle_8 = \gamma_4 \wedge \gamma_4 = 0 \tag{407}$$

$$\left\langle \gamma_2 \gamma_4 \right\rangle_6 = \gamma_2 \wedge \gamma_4 = 0 \tag{408}$$

$$\left\langle \gamma_4^2 \right\rangle_4 + \left\langle \gamma_2^2 \right\rangle_4 + \left(2\gamma_0 - 1\right)\gamma_4 = \left\langle \gamma_4^2 \right\rangle_4 + \gamma_2 \wedge \gamma_2 + \left(2\gamma_0 - 1\right)\gamma_4 = 0 \tag{409}$$

$$2\left\langle \gamma_2 \gamma_4 \right\rangle_2 + \left(2\gamma_0 - 1\right)\gamma_2 = 0 \tag{410}$$

$$\gamma_0\left(\gamma_0 - 1\right) I_{\mathcal{F}} + \left\langle \gamma_4^2 \right\rangle_0 + \left\langle \gamma_2^2 \right\rangle_0 = 0. \tag{411}$$

If one constrains

$$\gamma_0 = \frac{1}{2} \tag{412}$$

(thus possibly specifying a subset of solutions) then the idempotency conditions become

$$-\frac{1}{2} I_{\mathcal{F}} + \left\langle \gamma_2^2 \right\rangle_0 + \left\langle \gamma_4^2 \right\rangle_0 = 0 \tag{413}$$

$$2 \left\langle \gamma_2 \gamma_4 \right\rangle_2 = 0 \tag{414}$$

$$\left\langle \gamma_4^2 \right\rangle_4 + \gamma_2 \wedge \gamma_2 = 0 \tag{415}$$

$$\left\langle \gamma_2 \gamma_4 \right\rangle_6 = \gamma_2 \wedge \gamma_4 = 0 \tag{416}$$

$$\left\langle \gamma_4^2 \right\rangle_8 = \gamma_4 \wedge \gamma_4 = 0, \tag{417}$$

leading to

1.

$$-\frac{1}{2} I_{\mathcal{F}} + \left\langle \sum_{1 \le j,k \le r} c_j c_k y_j y_k y_{j+r} y_{k+r} \right\rangle_0 + \left\langle \sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le k_1 < k_2 < k_3 < k_4 \le 2r}} \gamma_{j_1 j_2 j_3 j_4} \gamma_{k_1 k_2 k_3 k_4} y_{j_1} y_{j_2} y_{j_3} y_{j_4} y_{k_1} y_{k_2} y_{k_3} y_{k_4} \right\rangle_0 \tag{418}$$

$$= \left\{ -\frac{1}{2} + \sum_{1 \le j \le r} c_j^2 + \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} \gamma_{j_1 j_2 j_3 j_4}^2 \right\} I_{\mathcal{F}} = 0, \tag{419}$$

i.e.

$$\sum_{1 \le j \le r} c_j^2 + \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} \gamma_{j_1 j_2 j_3 j_4}^2 = \frac{1}{2}. \tag{420}$$

2.

$$\left\langle \gamma_2 \gamma_4 \right\rangle_2 = \left\langle i \sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le j \le r}} \gamma_{j_1 j_2 j_3 j_4} c_j \left(y_j \wedge y_{j+r} \right) \left(y_{j_1} \wedge y_{j_2} \wedge y_{j_3} \wedge y_{j_4} \right) \right\rangle_2 \tag{421}$$

$$= i \sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le j \le r}} \gamma_{j_1 j_2 j_3 j_4} c_j \left\langle \left(y_j \wedge y_{j+r} \right) \left(y_{j_1} \wedge y_{j_2} \wedge y_{j_3} \wedge y_{j_4} \right) \right\rangle_2 = 0 \tag{422}$$

3.

$$\left\langle \sum_{\substack{1\le j_1<j_2<j_3<j_4\le 2r\\ 1\le k_1<k_2<k_3<k_4\le 2r}} \gamma_{j_1j_2j_3j_4}\gamma_{k_1k_2k_3k_4}\left(y_{j_1}\wedge y_{j_2}\wedge y_{j_3}\wedge y_{j_4}\right)\left(y_{k_1}\wedge y_{k_2}\wedge y_{k_3}\wedge y_{k_4}\right)\right\rangle_4 \tag{423}$$

$$+2\sum_{1\le j<k\le r} c_j c_k y_j\wedge y_k\wedge y_{j+r}\wedge y_{k+r} \tag{424}$$

$$= \sum_{\substack{1\le j_1<j_2<j_3<j_4\le 2r\\ 1\le k_1<k_2<k_3<k_4\le 2r}} \gamma_{j_1j_2j_3j_4}\gamma_{k_1k_2k_3k_4}\left\langle\left(y_{j_1}\wedge y_{j_2}\wedge y_{j_3}\wedge y_{j_4}\right)\left(y_{k_1}\wedge y_{k_2}\wedge y_{k_3}\wedge y_{k_4}\right)\right\rangle_4 \tag{425}$$

$$+2\sum_{1\le j<k\le r} c_j c_k y_j\wedge y_k\wedge y_{j+r}\wedge y_{k+r}=0 \tag{426}$$

4.

$$i\sum_{\substack{1\le j_1<j_2<j_3<j_4\le 2r\\ 1\le j\le r}} \gamma_{j_1j_2j_3j_4} c_j y_j\wedge y_{j+r}\wedge y_{j_1}\wedge y_{j_2}\wedge y_{j_3}\wedge y_{j_4}=0 \tag{427}$$

5.

$$\sum_{\substack{1\le j_1<j_2<j_3<j_4\le 2r\\ 1\le k_1<k_2<k_3<k_4\le 2r}} \gamma_{j_1j_2j_3j_4}\gamma_{k_1k_2k_3k_4} y_{j_1}\wedge y_{j_2}\wedge y_{j_3}\wedge y_{j_4}\wedge y_{k_1}\wedge y_{k_2}\wedge y_{k_3}\wedge y_{k_4}=0 \tag{428}$$

1. Special Case $\gamma_2 = 0$

Then the conditions become

$$-\frac{1}{2}I_{\mathcal{F}}+\left\langle\gamma_4^2\right\rangle_0=0\Leftrightarrow\left\langle\gamma_4^2\right\rangle_0=\frac{1}{2}I_{\mathcal{F}} \tag{429}$$

$$\left\langle\gamma_4^2\right\rangle_4=0 \tag{430}$$

$$\left\langle\gamma_4^2\right\rangle_8=\gamma_4\wedge\gamma_4=0, \tag{431}$$

The condition $\langle \gamma_4^2 \rangle_4 = 0$ gives

$$\left\langle \sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le k_1 < k_2 < k_3 < k_4 \le 2r}} \gamma_{j_1 j_2 j_3 j_4} \gamma_{k_1 k_2 k_3 k_4} \left(y_{j_1} \wedge y_{j_2} \wedge y_{j_3} \wedge y_{j_4}\right) \left(y_{k_1} \wedge y_{k_2} \wedge y_{k_3} \wedge y_{k_4}\right) \right\rangle_4 = \tag{432}$$

$$\sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le k_1 < k_2 < k_3 < k_4 \le 2r}} \gamma_{j_1 j_2 j_3 j_4} \gamma_{j_1 j_2 k_3 k_4} \left(y_{j_1} \wedge y_{j_2} \wedge y_{j_3} \wedge y_{j_4}\right) \left(y_{k_1} \wedge y_{k_2} \wedge y_{k_3} \wedge y_{k_4}\right) \tag{433}$$

$$+ \tag{434}$$

$$= 0 \tag{435}$$

$$A_4 \tag{436}$$

$$= A_{41234} \left|1234\right\rangle + A_{41235} \left|1235\right\rangle + A_{41236} \left|1236\right\rangle + A_{41245} \left|1245\right\rangle + A_{41246} \left|1246\right\rangle \tag{437}$$

$$+ A_{41256} \left|1256\right\rangle + A_{41345} \left|1345\right\rangle + A_{41346} \left|1346\right\rangle + A_{41356} \left|1356\right\rangle + A_{41456} \left|1456\right\rangle \tag{438}$$

$$+ A_{42345} \left|2345\right\rangle + A_{41246} \left|2346\right\rangle + A_{42356} \left|2356\right\rangle + A_{42456} \left|2456\right\rangle + A_{43456} \left|3456\right\rangle . \tag{439}$$

$$\left\langle A_4^2 \right\rangle_{(4)3456} = -A_{1234}A_{1256} + A_{1235}A_{1246} - A_{1236}A_{1245} - A_{1245}A_{1236} + A_{1246}A_{1235} - A_{1256}A_{1234} \tag{440}$$

$$= 2\left(A_{1235}A_{1246} - A_{1234}A_{1256} - A_{1236}A_{1245}\right) \tag{441}$$

$$\left\langle A_4^2 \right\rangle_{(4)1234} = 2\left(A_{5635}A_{1246} - A_{1234}A_{1256} - A_{1236}A_{1245}\right) \tag{442}$$

$$1234, 1235, 1236, 1245, 1246, 1256, 1345, 1346, 1356, 1456, 2345, 2346, 2356, 2456, 3456 \tag{443}$$

VI. FOCK-CLIFFORD CORRESPONDENCE.

⌈Let $r = 2$ then $\mathcal{F}$ is spanned by

$$I_{\mathcal{F}}, \left\{a_1^\dagger, a_2^\dagger\right\}, \left\{a_1, a_2\right\}, \left\{a_1^\dagger a_2^\dagger\right\}, \left\{a_1 a_2\right\}, \left\{a_1^\dagger a_2, a_2^\dagger a_1, a_1^\dagger a_1, a_2^\dagger a_2\right\}, \left\{a_1^\dagger a_2^\dagger a_1, a_1^\dagger a_2^\dagger a_2\right\}, \left\{a_1^\dagger a_1 a_2, a_2^\dagger a_1 a_2\right\}, \left\{a_1^\dagger a_2^\dagger \tag{444}$$

$$\dim\left\{\mathcal{A}_{00} \oplus \mathcal{A}_{10} \oplus \mathcal{A}_{01} \oplus \mathcal{A}_{20} \oplus \mathcal{A}_{02} \oplus \mathcal{A}_{11} \oplus \mathcal{A}_{21} \oplus \mathcal{A}_{12} \oplus \mathcal{A}_{22}\right\} \tag{445}$$

$$= \binom{r}{0}^2 + 2\binom{r}{0}\binom{r}{2} + 2\binom{r}{0}\binom{r}{1} + \binom{r}{1}^2 + 2\binom{r}{1}\binom{r}{2} + \binom{r}{2}^2 = 1 + 4 + 4 + 2 + 4 + 1 = 16 = 2^4 \tag{446}$$

$$\dim\left\{\mathcal{A}_{00} \oplus \mathcal{A}_{11} \oplus \mathcal{A}_{22}\right\} = \binom{r}{0}^2 + \binom{2}{1}^2 + \binom{2}{2}^2 = 1 + 4 + 1 = 6 \tag{447}$$

and by

$$I_{\mathcal{F}}, \{x_1, x_2, x_3, x_4\}, \{x_1x_2, x_1x_3, x_1x_4, x_2x_3, x_2x_4, x_3x_4\}, \{x_1x_2x_3, x_1x_2x_4, x_1x_3x_4\}, \{x_1x_2x_3x_4\} \tag{448}$$

$$\dim\{\mathcal{F}_0 \oplus \mathcal{F}_1 \oplus \mathcal{F}_2 \oplus \mathcal{F}_3 \oplus \mathcal{F}_4\} = \binom{2r}{0} + \binom{2r}{1} + \binom{2r}{2} + \binom{2r}{3} + \binom{2r}{4} = 1 + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} \tag{449}$$

$$= 1 + 4 + 6 + 4 + 1 = 16 = 2^4 \tag{450}$$

$$\dim\{\mathcal{F}_0 \oplus \mathcal{F}_2 \oplus \mathcal{F}_4\} = \binom{2r}{0} + \binom{2r}{2} + \binom{2r}{4} = \binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 1 + 6 + 1 = 8 = 2^3 \tag{451}$$

$$\mathcal{F}_0 \oplus \mathcal{F}_2 \oplus \mathcal{F}_4 \supset \mathcal{A}_{00} \oplus \mathcal{A}_{11} \oplus \mathcal{A}_{22} \tag{452}$$

Let $r=3$ then $\mathcal{F}$ is spanned by

$$I_{\mathcal{F}}, \left\{a_1^\dagger, a_2^\dagger, a_3^\dagger\right\}, \{a_1, a_2, a_3\}, \left\{a_1^\dagger a_2^\dagger, a_1^\dagger a_3^\dagger, a_2^\dagger a_3^\dagger\right\}, \left\{a_1^\dagger a_2^\dagger a_3^\dagger\right\}, \{a_1a_2, a_1a_3, a_2a_3\}, \{a_1a_2a_3\}, \tag{453}$$

$$\left\{a_1^\dagger a_2, a_2^\dagger a_1, a_1^\dagger a_3, a_3^\dagger a_1, a_2^\dagger a_3, a_3^\dagger a_2, a_1^\dagger a_1, a_2^\dagger a_2, a_3^\dagger a_3\right\}, \tag{454}$$

$$\left\{a_1^\dagger a_2^\dagger a_1, a_1^\dagger a_2^\dagger a_2, a_1^\dagger a_2^\dagger a_3, a_1^\dagger a_3^\dagger a_1, a_1^\dagger a_3^\dagger a_2, a_1^\dagger a_3^\dagger a_3, a_2^\dagger a_3^\dagger a_1, a_2^\dagger a_3^\dagger a_2, a_2^\dagger a_3^\dagger a_3\right\}, \tag{455}$$

$$\left\{a_1^\dagger a_1a_2, a_2^\dagger a_1a_2, a_3^\dagger a_1a_2, a_1^\dagger a_1a_3, a_2^\dagger a_1a_3, a_3^\dagger a_1a_3, a_1^\dagger a_2a_3, a_2^\dagger a_2a_3, a_3^\dagger a_2a_3\right\}, \tag{456}$$

$$\left\{a_1^\dagger a_2^\dagger a_2a_1, a_1^\dagger a_3^\dagger a_3a_1, a_2^\dagger a_3^\dagger a_3a_2, a_1^\dagger a_2^\dagger a_3a_1, a_1^\dagger a_2^\dagger a_3a_2, a_1^\dagger a_3^\dagger a_2a_1, a_1^\dagger a_3^\dagger a_3a_2, a_2^\dagger a_3^\dagger a_2a_1, a_2^\dagger a_3^\dagger a_3a_1\right\}, \tag{457}$$

$$\left\{a_1^\dagger a_2^\dagger a_3^\dagger\right\}, \{a_1a_2a_3\}, \left\{a_1^\dagger a_2^\dagger a_3^\dagger a_1, a_1^\dagger a_2^\dagger a_3^\dagger a_2, a_1^\dagger a_2^\dagger a_3^\dagger a_3\right\}, \left\{a_1^\dagger a_1a_2a_3, a_2^\dagger a_1a_2a_3, a_2^\dagger a_1a_2a_3\right\}, \tag{458}$$

$$\left\{a_1^\dagger a_2^\dagger a_3^\dagger a_1a_2, a_1^\dagger a_2^\dagger a_3^\dagger a_1a_3, a_1^\dagger a_2^\dagger a_3^\dagger a_2a_3\right\}, \left\{a_1^\dagger a_2^\dagger a_1a_2a_3, a_1^\dagger a_3^\dagger a_1a_2a_3, a_2^\dagger a_3^\dagger a_1a_2a_3\right\}, \left\{a_1^\dagger a_2^\dagger a_3^\dagger a_1a_2a_3\right\} \tag{459}$$

$$\dim\left\{\begin{array}{c}\mathcal{A}_{00}\oplus(\mathcal{A}_{10}\oplus\mathcal{A}_{01})\oplus\mathcal{A}_{11}\oplus(\mathcal{A}_{20}\oplus\mathcal{A}_{02})\oplus(\mathcal{A}_{21}\oplus\mathcal{A}_{12})\oplus\mathcal{A}_{22}\\ \oplus(\mathcal{A}_{30}\oplus\mathcal{A}_{03})\oplus(\mathcal{A}_{31}\oplus\mathcal{A}_{13})\oplus(\mathcal{A}_{23}\oplus\mathcal{A}_{32})\oplus\mathcal{A}_{33}\end{array}\right\} \tag{460}$$

$$=\binom{r}{0}^2+2\binom{r}{0}\binom{r}{1}+2\binom{r}{0}\binom{r}{2}+2\binom{r}{0}\binom{r}{3}+\binom{r}{1}^2+2\binom{r}{1}\binom{r}{2} \tag{461}$$

$$+2\binom{r}{1}\binom{r}{3}+\binom{r}{2}^2+2\binom{r}{2}\binom{r}{3}+\binom{r}{3}^2 \tag{462}$$

$$=\left(\binom{r}{0}+\binom{r}{1}+\binom{r}{2}+\binom{r}{3}\right)\left(\binom{r}{0}+\binom{r}{1}+\binom{r}{2}+\binom{r}{3}\right)=(2^r)^2 \tag{463}$$

$$=\left(\binom{3}{0}+\binom{3}{1}+\binom{3}{2}+\binom{3}{3}\right)\left(\binom{3}{0}+\binom{3}{1}+\binom{3}{2}+\binom{3}{3}\right) \tag{464}$$

$$=(1+3+3+1)^2=64=2^6 \tag{465}$$

$$=\binom{3}{0}^2+2\binom{3}{0}\binom{3}{1}+2\binom{3}{0}\binom{3}{2}+2\binom{3}{0}\binom{3}{3}+\binom{3}{1}^2+2\binom{3}{1}\binom{3}{2} \tag{466}$$

$$+2\binom{3}{1}\binom{3}{3}+\binom{3}{2}^2+2\binom{3}{2}\binom{3}{3}+\binom{3}{3}^2 \tag{467}$$

$$=1+6+6+2+9+18+6+9+6+1=64=2^6 \tag{468}$$

$$\dim\{\mathcal{A}_{00}\oplus\mathcal{A}_{11}\oplus\mathcal{A}_{22}\oplus\mathcal{A}_{33}\}=\binom{3}{0}^2+\binom{3}{1}^2+\binom{3}{2}^2+\binom{3}{3}^2=1+9+9+1=20 \tag{469}$$

and by

$$I_{\mathcal{F}}, \{x_1, x_2, x_3, x_4, x_5, x_6\}, \{x_1x_2, x_1x_3, x_1x_4, , x_1x_5, x_1x_6, x_2x_3, x_2x_4, x_2x_5, x_2x_6, x_3x_4, x_3x_5, x_3x_6, x_4x_5, x_4x_6, x \tag{470}$$

$$\left\{ \begin{array}{c} x_1x_2x_3, x_1x_2x_4, x_1x_2x_5, x_1x_2x_6, x_1x_2x_4, x_1x_2x_5, x_1x_2x_6, x_1x_3x_4, x_1x_3x_5, x_1x_3x_6, x_1x_4x_5, \\ x_1x_4x_6, x_1x_5x_6, x_2x_3x_4, x_2x_3x_5, x_2x_3x_6, x_2x_4x_5, x_2x_4x_6, x_2x_5x_6, x_3x_4x_5, x_3x_5x_6, x_4x_5x_6 \end{array} \right\}, \tag{471}$$

$$\left\{ \begin{array}{c} x_1x_2x_3x_4, x_1x_2x_3x_5, x_1x_2x_3x_6, x_1x_2x_4x_5, x_1x_2x_4x_6, x_1x_2x_5x_6, x_1x_3x_4x_5, x_1x_3x_4x_6, x_1x_3x_5x_6, \\ x_1x_4x_5x_6, x_2x_3x_4x_5, x_2x_3x_4x_6, x_2x_4x_5x_6, x_2x_3x_5x_6, x_3x_4x_5x_6 \end{array} \right\}, \tag{472}$$

$$etc..... \tag{473}$$

$$\dim\{\mathcal{F}_0 \oplus \mathcal{F}_1 \oplus \mathcal{F}_2 \oplus \mathcal{F}_3 \oplus \mathcal{F}_4 \oplus \mathcal{F}_5 \oplus \mathcal{F}_6\} = \binom{2r}{0} + \binom{2r}{1} + \binom{2r}{2} + \binom{2r}{3} \tag{474}$$

$$= \binom{6}{0} + \binom{6}{1} + \binom{6}{2} + \binom{6}{3} + \binom{6}{4} + \binom{6}{5} + \binom{6}{6} \tag{475}$$

$$= 1 + 6 + 15 + 20 + 15 + 6 + 1 = 64 = 2^6 \tag{476}$$

$$\dim\{\mathcal{F}_0 \oplus \mathcal{F}_2 \oplus \mathcal{F}_4 \oplus \mathcal{F}_6\} \supset \mathcal{A}_{00} \oplus \mathcal{A}_{11} \oplus \mathcal{A}_{22} \oplus \mathcal{A}_{33} \tag{477}$$

$$= \binom{2r}{0} + \binom{2r}{2} + \binom{2r}{4} = 1 + \binom{6}{2} + \binom{6}{4} + \binom{6}{6} = 1 + 15 + 15 + 1 = 32 = 2^5 \tag{478}$$

$\rfloor$

In general

$$\mathcal{F} = \bigoplus_{0 \le j \le 2r} \mathcal{F}_j \tag{479}$$

$$\dim\{\mathcal{F}\} = \sum_{0 \le j \le 2r} \binom{2r}{j} = 2^{2r} \tag{480}$$

$$\mathcal{F}_+ = \bigoplus_{0 \le j \le r} \mathcal{F}_{2j} \tag{481}$$

$$\dim\{\mathcal{F}_+\} = \sum_{0 \le j \le r} \binom{2r}{2j} = 2^{2r-1}, \tag{482}$$

and

$$\mathcal{F}= \bigoplus_{0\leq j,k\leq r} \mathcal{A}_{jk} \tag{483}$$

$$\dim\{\mathcal{F}\} = \sum_{0\leq j,k\leq r} \binom{r}{j}\binom{r}{k} = \left(\sum_{0\leq j\leq r} \binom{r}{j}\right)^2 = (2^r)^2 = 2^{2r} \tag{484}$$

$$\mathcal{F}_+ \supset \mathcal{F}_{con} = \bigoplus_{0\leq j\leq r} \mathcal{A}_{jj} \tag{485}$$

$$\dim\{\mathcal{F}_{con}\} = \sum_{0\leq j\leq r} \binom{r}{j}^2. \tag{486}$$

The $\mathcal{A}$ filtration is finer than the $\mathcal{F}$ filtration. $\mathcal{F}_+$ and $\mathcal{F}_{con}$ are subalgebras of $\mathcal{F}$.

VII. SPIN OPERATORS

$$S_0 = \frac{1}{2}\sum_{1\leq j\leq s} \left\{a_j^\dagger a_j - a_{j+s}^\dagger a_{j+s}\right\} \tag{487}$$

$$S_{+1} = \sum_{1\leq j\leq s} a_j^\dagger a_{j+s} \tag{488}$$

$$S_{-1} = \sum_{1\leq j\leq s} a_{j+s}^\dagger a_j \tag{489}$$

$$S^2 = S_{-1}S_{+1} + S_0\left(S_0 + I_{\mathcal{F}}\right), \tag{490}$$

where

$$\begin{aligned} S_{+1}\left|\varphi_k\right\rangle &= 0 \\ S_{+1}\left|\varphi_{k+s}\right\rangle &= \left|\varphi_k\right\rangle \\ S_{-1}\left|\varphi_k\right\rangle &= \left|\varphi_{k+s}\right\rangle \\ S_{-1}\left|\varphi_{k+s}\right\rangle &= 0 \\ S_0\left|\varphi_k\right\rangle &= \frac{1}{2}\left|\varphi_k\right\rangle \\ S_0\left|\varphi_{k+s}\right\rangle &= -\frac{1}{2}\left|\varphi_{k+s}\right\rangle \\ 1 \leq k &\leq s. \end{aligned} \tag{491}$$